

Introduction to Mathematics and Statistics for Steam_AI Classes

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**Lesson: Innovative AI and Probability Teaching
Methods for Middle School STEM**

Contents

What is Probability?	8
Calculating Probability.....	8
Example 1:.....	9
Example 2:.....	10
What is Probability Space?	11
Example 1: Dice Rolling.....	12
Example 2:.....	13
The Importance of Probability	14
Example 1: Supplier Inventory Management.....	14
Example 2: Stem Robot Kits.....	15
Example 3: Robot Competition.....	16
Situation:	16
Historical Data:	16
Role of Probability:	16
Probability Calculations:	16
Decision-Making Process:.....	16
Example 4: Plant Growing Competition	17
Historical Data:	17
Use of Probability:	17
Probability Calculations:	17
Decision-Making Process:.....	17
Key Terms in Probability	18
Fun Examples.....	19
Example 1: Candy Bag.....	19
Relationship Between Events.....	20
1. Mutually Exclusive Events.....	20
2. Mutually Inclusive Events.....	20
Fun Examples.....	21
Example 1: Fruit Selection.....	21
Example 2: Classroom Tests	21
STEM Project Example.....	22
Probability Approach	23

1. Classical Approach.....	23
2. Relative Frequency Approach.....	23
3. Subjective Approach	23
What Does Probability Mean?	23
Basic Terms of Probability.....	23
Examples of the Importance of Probability.....	24
Line-Following Robot and Probability	25
Application of Probability and Conditional Probability Sensor Data and Probabilities ..	25
Conditional Probability Calculations	25
Example Calculation	25
Fuzzy Logic and Probability	26
Conclusion.....	26
Autonomous Drone Design and Probability	27
Application of Probability and Conditional Probability Sensor Data and Probabilities ..	27
Fuzzy Logic and Probability	28
Summary	28
Independent and Dependent Events	29
Independent Events	29
Dependent Events.....	30
Example: Card Drawing.....	30
Summary	31
Application for STEM Students: Autonomous Drone	32
Example: Sensor Data and Independent/Dependent Events	32
Line-Following Robot and Events	33
Example Calculation:	33
The Complement of an Event.....	34
Basic Concept.....	34
Explanation with Examples.....	34
Example 1: Dice Rolling	34
Example 2: Coin Tossing	34
Example 3: Line-Following Robot.....	35
Example 4: Classroom Test Results.....	36
Example 5: Weather Forecast	36

Probability Addition Theorem	37
Basic Concepts	37
Mutually Exclusive Events.....	37
Non-Mutually Exclusive Events	37
Addition Theorem	37
Explanation with Examples.....	37
Example 1: Mutually Exclusive Events	37
Example 2: Non-Mutually Exclusive Events	38
Application for STEM Students: Line-Following Robot	39
Product Probability Theorem	40
Independent Events	40
Dependent Events.....	40
Explanation with Examples.....	40
Example: Electronic System Components	41
Application for STEM Students: Line-Following Robot Situation:	43
Sensor Operation	43
What is Bayes' Theorem?.....	44
Bayes' Theorem Formula	44
Example: Success of a New Product in the Market	44
Updating Probabilities Using Bayes' Theorem	44
Step 1: Calculate the Probability of B	44
Step 2: Calculate Updated Probabilities	45
Application for STEM Students: Line-Following Robot.....	46
Scenario: Autonomous Drone and Bayes' Theorem	47
What is a Random Variable?.....	49
Types of Random Variables.....	49
Example: Rolling Three Dice.....	49
Real-Life Examples.....	50
Application for STEM Students: Line-Following Robot.....	51
Types of Random Variables.....	52
Discrete Random Variable	52
Continuous Random Variable	52
Explanation with Examples.....	53

Discrete Random Variable	53
Continuous Random Variable	53
Application for STEM Students: Line-Following Robot	54
What is a Probability Distribution?	55
Elements of a Probability Distribution	55
Visualization of Probability Distributions	55
Example: Perfect and Defective Parts in a Production Process	55
Probability Distribution with Examples	56
Example 1: Rolling a Die	56
Example 2: Product Sales	56
Application for STEM Students: Autonomous Drone	57
What is Expected Value?	59
Calculating Expected Value	59
Example: Expected Value of Rolling a Die	59
Probability Distribution and Expected Value	60
Example 1: Number of Customers in a Theater	60
Example 2: Defective Parts in a Production Process	61
Application for STEM Students: Autonomous Drone	62
Random Variable and Probability Distribution	63
Application: Mechanical Hand Tool Experiment	63
Discrete Probability Distribution	66
Bernoulli Distribution	66
Binomial Distribution	66
Poisson Distribution	67
Summary	67
Continuous Probability Distribution	68
Normal Distribution	68
Example: Student Grades	68
Uniform Distribution	71
Summary	72
Normal Probability Distribution	73
Parameters of the Normal Distribution	73
Example: Normal Probability Distribution	74

Example: Student Heights	75
Example: Balloon Height Competition	78
Uniform Probability Distribution	81
Parameters of Uniform Distribution.....	81
Characteristics of Uniform Distribution.....	81
Uniform Probability Distribution Equation.....	81
Uniform Probability Distribution: Examples	81
Example: Random Number Generation in Games	84

Story

"The STEM class preparing for the International Robo-Taxi Autonomous Riding Competition is planning to add new features to their vehicle. These new features could enhance the vehicle's performance and help achieve higher scores in the competition. However, these innovations also carry some risks. Developing and integrating new features can be costly, and the anticipated performance improvements may not be realized. Are these changes a wise move? What is the likelihood of success?"

What is Probability?

Probability is the chance or likelihood of an event occurring. It expresses the possibility of something happening or not happening. Example Events:

- **Team Performance:** The probability of a sports team winning a match.
- **Time Spent Traveling:** The probability of completing a journey within a specific time frame.
- **Rainfall Level:** The probability of rain tomorrow.
- **Availability of Products:** The probability of finding a desired product in the store.

Calculating Probability

To calculate probability, we divide the number of favorable outcomes by the total number of possible outcomes. Probability Formula:

Probability = Favorable Outcomes / Total Possible Outcomes

Example 1:

Let's say there are 5 red, 3 blue, and 2 green balls in a bag. We want to find the probability of drawing a red ball when we randomly pick one.

Step 1: Favorable Outcomes

- Number of red balls: 5

Step 2: Total Possible Outcomes

- Total number of balls: 5 red + 3 blue + 2 green = 10 balls

Step 3: Calculating

- Probability Probability of drawing a red ball = $5/10 = 0.5$

This means there is a 50% chance of drawing a red ball.

Example 2:

There are 20 students in a class, and the teacher will select one student to give candy to. There are 12 female students and 8 male students in the class. Now, let's calculate the probability of the teacher selecting a female student.

Step 1: Favorable Outcomes

- Number of female students: 12

Step 2: Total Possible Outcomes

- Total number of students: 20

Step 3: Calculating Probability

- Probability of selecting a female student = $12/20 = 0.6$

This means there is a 60% chance of selecting a female student. This way, we can explain the concept of probability in a fun and understandable manner.

What is Probability Space?

Probability space is a mathematical structure that helps us understand the likelihood of an event occurring. It has three important components:

1. **Sample Space (Ω):** Contains all possible outcomes of an event.
2. **Events (A):** Specific outcomes or combinations of outcomes selected from the sample space.
3. **Probability Function (P):** Assigns a numerical value to each event, which represents the probability of the event occurring.

Example 1: Dice Rolling

Let's consider a dice-rolling experiment. A die has 6 faces, each bearing numbers from 1 to 6. We'll use this experiment to explain the probability space.

1. **Sample Space (Ω)** The sample space includes all possible outcomes that can result from rolling the die. We consider which numbers could appear on the top face of the die:

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

2. **Events (A)** An event consists of specific outcomes within the sample space. For example:

- **Event A:** The die shows a 2. $A = \{2\}$
- **Event B:** The die shows an even number (2, 4, 6)

3. **Probability Function (P)** The probability function assigns a numerical value to each event. Since each face of the die is equally likely to appear, the probability for each face is $1/6$.

- **Probability for Event A:** $P(A) = 1/6$
- **Probability for Event B:** The probability of getting an even number (with 3 possible outcomes): $P(B) = 3/6 = 1/2$

Example 2:

Imagine a bag containing 3 red, 2 blue, and 1 green ball. We will randomly draw a ball from this bag.

1. **Sample Space (Ω)** The colors of all the balls in the bag:

$$\Omega = \{\text{red, red, red, blue, blue, green}\}$$

2. **Events (A)**

- **Event A:** The event of drawing a red ball.

$$A = \{\text{red, red, red}\}$$

3. **Probability Function (P)** The probability of drawing each ball:

- **Probability for Event A:**

$$P(A) = \text{number of red balls} / \text{total number of balls} = 3/6 = 1/2$$

The Importance of Probability

Probability is a powerful tool used to handle uncertainties in planning and decision-making processes. It helps us assess and predict outcomes, prepare, and take action when circumstances and factors change.

Example 1: Supplier Inventory Management

Situation:

A supplier must decide the optimum amount of inventory to keep on hand to prevent stockouts. The goal is to serve customers effectively and prevent them from turning to another supplier due to lack of inventory.

Historical Data:

Past sales data indicate an average daily sales volume of 80 units. However, this sales quantity can vary daily. Some days might see 70 units sold, while other days might see 90 units sold.

Role of Probability:

Probability can help the supplier statistically determine the appropriate number of units to keep in stock. How is it used?

1. Analysis of Past Data:

- The supplier analyzes past data and determines that the average daily sales are 80 units.

2. Examination of Variability Factors:

- The supplier observes that daily sales fluctuate between 70 and 90 units.

3. Creation of a Probability Distribution:

- The supplier assumes that sales follow a normal distribution, indicating that sales will mostly be around 80 units but can sometimes be lower or higher.

4. Probability Calculations:

- The supplier uses probability functions to calculate the likelihood of selling a certain number of units on a given day.
- For example, the probability of selling 85 units might be 30%, and the probability of selling 75 units might be 25%.

5. Determination of Optimum Stock Quantity:

- The supplier determines the minimum number of units to keep in stock to meet customer demands. Let's say they want at least a 95% probability that they have enough stock.
- In this case, they calculate that maintaining a stock of 85 units will be sufficient with a 95% probability.

Example 2: Stem Robot Kits

A school is purchasing robot kits to participate in a STEM competition. In previous years, the number of students participating varied between 50 and 70. The school wants to stock enough robot kits but does not want to exceed the budget by stocking too much. Historical Data:

Based on past data, an average of 60 students participate. Use of Probability:

1. Analysis of Past Data:

- The school analyzes the number of students who participated in previous years and determines an average of 60.

2. Examination of Variability Factors:

- It observes that the number of participants varies between 50 and 70.

3. Creation of a Probability Distribution:

- The school assumes that the number of students follows a normal distribution.

4. Probability Calculations:

- The school calculates the probability of a certain number of students participating.
- For example, the probability of 65 students participating might be 40%, and the probability of 55 students participating might be 20%.

5. Determination of Optimum Kit Quantity:

- The school wants to have enough robot kits with a 95% probability.
- In this case, it calculates that stocking 65 robot kits will be sufficient with a 95% probability.

This example illustrates how probability is used in planning and decision-making processes. Probability is a powerful tool for managing uncertainty and making the best decisions.

Example 3: Robot Competition

Situation:

The STEM class is preparing to participate in an international robot competition. In this competition, the robots are required to perform specific tasks. However, it is uncertain whether the robots will successfully complete these tasks. The STEM class plans to add new sensors to their robots. These sensors could help the robots complete the tasks more accurately and quickly. However, adding sensors also carries some risks. The students want to evaluate the likelihood of success from adding these sensors.

Historical Data:

In past competitions, robots without sensors had a 60% task completion rate. Robots with sensors had an 80% completion rate.

Role of Probability:

Probability can help students evaluate the success of adding sensors and make decisions. How it is used?

1. Analysis of Past Data:

- Students determine that robots without sensors were successful 60% of the time.
- Robots with sensors were successful 80% of the time.

2. Decision Making on Adding Sensors:

- Students perform probability calculations to decide whether to add sensors to their robots.

Probability Calculations:

Scenario 1: Robots Without Sensors:

- Probability of success for a robot without sensors:
 $P(\text{Success}) = 0.60$
- Probability of failure for a robot without sensors:
 $P(\text{Failure}) = 1 - P(\text{Success}) = 1 - 0.60 = 0.40$

Scenario 2: Robots With Sensors:

- Probability of success for a robot with sensors:
 $P(\text{Success}) = 0.80$
- Probability of failure for a robot with sensors:
 $P(\text{Failure}) = 1 - P(\text{Success}) = 1 - 0.80 = 0.20$

Decision-Making Process:

The students see that adding sensors increases the chance of success for the robot. However, they must also consider the cost of the sensors and the effort required to install them. With this information, the students realize they could achieve a higher chance of success by adding sensors.

Example 4: Plant Growing Competition

A STEM class decides to participate in a plant growing competition. In the competition, students are expected to grow their plants as quickly and healthily as possible. The students want to try different watering methods for their plants.

Historical Data:

In past competitions, students who used manual watering achieved a 50% success rate. Students who used an automatic watering system achieved a 90% success rate.

Use of Probability:

1. Analysis of Past Data:

- They determine that plants watered manually were successful 50% of the time.
- Plants watered using an automatic system were successful 90% of the time.

2. Decision to Use Automatic Watering System:

- Students perform probability calculations to decide to water their plants with an automatic system.

Probability Calculations:

Scenario 1: Manual Watering:

- Probability of success for a student's plant with manual watering:

$$P(\text{Success}) = 0.50$$

- Probability of failure for a student's plant with manual watering:

$$P(\text{Failure}) = 1 - P(\text{Success}) = 1 - 0.50 = 0.50$$

Scenario 2: Automatic Watering:

- Probability of success for a student's plant using an automatic system:

$$P(\text{Success}) = 0.90$$

- Probability of failure for a student's plant using an automatic system:

$$P(\text{Failure}) = 1 - P(\text{Success}) = 1 - 0.90 = 0.10$$

Decision-Making Process:

The students see that using an automatic watering system increases the likelihood that their plants will grow healthier and faster. This increases their chances of success in the competition. However, they also consider the cost and effort required to set up the automatic system. These examples demonstrate how probability plays a significant role in STEM projects and how it handles uncertainties.

Key Terms in Probability

1. **Event** Events are specific situations considered in probability studies. Probabilities are assigned to these events. Example:
 - **Event:** Rolling an even number on a die.
 - **Explanation:** Getting a 2, 4, or 6 on the die is an event. In this case, the probability of rolling an even number is calculated.
2. **Experiment** An experiment is a planned activity conducted in a controlled environment. These activities are performed to understand the probability of certain events occurring. Example:
 - **Experiment:** Rolling a die.
 - **Explanation:** The action of rolling the die and observing which number comes up.
3. **Outcome** An outcome is the result of an experiment. Each experiment can have various possible outcomes. Example:
 - **Outcome:** Rolling a 3 on a die.
 - **Explanation:** The number 3 appearing on the top face of the die after it is rolled is an outcome.
4. **Sample Space** The sample space is a list of all possible outcomes of an experiment. Example:
 - **Sample Space:** All possible outcomes when a die is rolled: {1, 2, 3, 4, 5, 6}
 - **Explanation:** All numbers that can appear when the die is rolled constitute the sample space.
5. **Probability Function** The probability function describes the likelihood of an outcome occurring. Example:
 - **Probability Function:** The probability of rolling an even number when a die is rolled. $P(\text{Even number}) = \text{Number of even numbers} / \text{Total number of possible outcomes} = 3 / 6 = 0.5$
 - **Explanation:** The probability of rolling an even number is 0.5 or 50%.
6. **Multiple Events** Often in studies or research, two or more events are considered simultaneously. Example:
 - **Event 1:** Rolling an even number on a die.
 - **Event 2:** Rolling a number greater than 3 on the same die roll.
 - **Explanation:** In this case, the probability of both rolling an even number and a number greater than 3 is considered (4 or 6).

Fun Examples

Example 1: Candy Bag

There are 5 red, 3 blue, and 2 green candies in a bag. We will randomly draw a candy.

- **Experiment:** Drawing a candy from the bag.
- **Outcome:** Drawing a red, blue, or green candy.
- **Sample Space:** {red, blue, green}
- **Event:** Drawing a red candy.
- **Probability Function:** The probability of drawing a red candy:
$$P(\text{Red}) = \text{Number of red candies} / \text{Total number of candies} = 5/10 = 0.5$$

Example 2: School Bus

There are 20 students on a school bus, of which 12 are girls and 8 are boys. We are selecting a student at random.

- **Experiment:** Selecting a random student from the bus.
- **Outcome:** Selecting a girl or boy.
- **Sample Space:** {girl, boy}
- **Event:** Selecting a girl.
- **Probability Function:** The probability of selecting a girl:
$$P(\text{Girl}) = 12/20 = 0.6$$

These examples use everyday scenarios from the lives of STEM students to help them more easily understand key terms in probability.

Relationship Between Events

In studies, the relationship between two or more events is examined. These relationships are primarily of two types:

1. **Mutually Exclusive Events**
2. **Mutually Inclusive Events**

1. Mutually Exclusive Events

Events are called mutually exclusive if they cannot occur at the same time. That is, if one event occurs, the other cannot. Example:

- **Event A:** Drawing a card that is a heart.
- **Event B:** Drawing a card that is a spade. Explanation: In a deck of cards, the same card cannot be both a heart and a spade. Therefore, these two events are mutually exclusive.

2. Mutually Inclusive Events

Events are mutually inclusive if at least one of the events must occur. That is, both events can occur at the same time or one can occur when the other does. Example:

- **Event A:** Rolling an odd number on a die.
- **Event B:** Rolling an even number on a die.

Explanation: When a die is rolled, it always results in either an odd or an even number. Therefore, these events are mutually inclusive because every roll of the die will necessarily result in either an odd or an even number.

Fun Examples

Example 1: Fruit Selection

There are apples and oranges in a bag. We randomly select a fruit.

- **Mutually Exclusive:**
 - **Event A:** The selected fruit being an apple.
 - **Event B:** The selected fruit being an orange.
- **Explanation:** A fruit cannot be both an apple and an orange at the same time. Therefore, these events are mutually exclusive.

Example 2: Classroom Tests

Math and science tests are being conducted in a classroom. Students can solve both tests.

- **Mutually Inclusive:**
 - **Event A:** The student solving the math test.
 - **Event B:** The student solving the science test.
- **Explanation:** A student can solve both the math and the science tests. Therefore, these events are mutually inclusive.

Understanding the Relationship Between Events

Knowing whether events are mutually exclusive or inclusive is important in probability calculations. This helps us understand which events can occur together and which cannot.

STEM Project Example

Project: A STEM class is participating in a robot competition. The robots need to accomplish two different tasks: reaching a target and avoiding obstacles.

Mutually Exclusive:

- **Event A:** The robot reaches the target.
- **Event B:** The robot fails to avoid obstacles and crashes.
- **Explanation:** If the robot reaches the target, it does not crash into obstacles. If the robot crashes into obstacles, it does not reach the target. These events are mutually exclusive.

Mutually Inclusive:

- **Event A:** The robot reaches the target.
- **Event B:** The robot avoids obstacles.
- **Explanation:** The robot can reach the target while simultaneously avoiding obstacles. These events are mutually inclusive because both can occur at the same time.

Probability Approach

Probability expresses the chance of a particular event occurring. There are three main approaches to probability: classical approach, relative frequency approach, and subjective approach.

1. Classical Approach

In the classical approach, probability is assumed where all possible outcomes are equally likely. Probability is calculated as the ratio of favorable outcomes to the total possible outcomes. Example:

- **Drawing a Card:** The probability of drawing a hearts card from a deck of 52 cards.
 $P(\text{Hearts}) = \text{Number of hearts cards} / \text{Total number of cards} = 13/52 = 1/4$
- **Explanation:** There are 52 cards in the deck and 13 are hearts cards. The probability of drawing a hearts card is 13/52, which is 1/4 or 25%.

2. Relative Frequency Approach

In the relative frequency approach, probabilities are determined using past data or experimental results. The ratio of favorable outcomes to total possible outcomes is considered. Example:

- **Defective Product:** It has been observed that 5 out of 1000 products manufactured in a factory are defective. $P(\text{Defective Product}) = \text{Number of defective products} / \text{Total products manufactured} = 5/1000$
- **Explanation:** Based on past data, 5 out of 1000 products are defective. Therefore, the probability of a product being defective is 0.005 or 0.5%.

3. Subjective Approach

In the subjective approach, probabilities are determined based on past experiences and expert judgments. This approach relies on expert opinions to estimate the likelihood of events occurring. Example:

- **Customer Behavior:** The probability of a customer purchasing a specific brand.
 - Experts estimate the probability of a customer buying Brand A at 70% based on past consumer behaviors and market research.
 - **Explanation:** The expert's subjective assessment is based on past experience and market data. Therefore, the probability of the customer buying Brand A is 70%.

What Does Probability Mean?

Probability plays a crucial role in decision-making as it provides a framework to assess and measure uncertainty in risk assessment, decision analysis, forecasting, and optimization.

Basic Terms of Probability

- **Event (Event):** Specific situations considered in probability studies.
- **Probability (Probability):** The chance of an event occurring.

Examples of the Importance of Probability

Example 1: Classroom Test

- **Experiment:** Randomly selecting a student from the class.
- **Outcome:** Selecting a girl or boy student.
- **Sample Space:** {girl, boy}
- **Probability Function:** The probability of selecting a girl. $P(\text{Girl}) = \frac{\text{Number of girl students}}{\text{Total number of students}} = \frac{12}{20} = 0.6$
- **Explanation:** There are 20 students in the class, 12 of whom are girls. The probability of selecting a girl is 0.6 or 60%.

Example 2: Factory Production

- **Relative Frequency Approach:** We know that out of 1000 products manufactured in the factory, 10 are defective. $P(\text{Defective Product}) = \frac{10}{1000} = 0.01$
- **Explanation:** Based on past data, 1% of the products are defective.

Example 3: Expert Estimation

- **Subjective Approach:** Experts estimate the probability of success for a new product in the market at 80% based on market trends.
- **Explanation:** The market knowledge and experience of experts have been used in making this estimate.

Line-Following Robot and Probability

The primary goal of a line-following robot is to follow a black line drawn on the ground. The robot is equipped with various sensors that detect the position of the line. Based on the information from these sensors, adjustments are made to the robot's direction and speed.

Application of Probability and Conditional Probability Sensor Data and Probabilities

Let's assume that the robot has two sensors, one on the right and one on the left of the line. These sensors return a value of 1 when they detect the black line and 0 when they do not.

- **Sensor A (Left Sensor):** Returns 1 if the left sensor detects the line, otherwise 0.
- **Sensor B (Right Sensor):** Returns 1 if the right sensor detects the line, otherwise 0.

Conditional Probability Calculations

Conditional probability can be used to decide which direction the robot should move based on the information from the sensors. Situation:

- **Event A:** The robot turning right (Line on the right).
- **Event B:** The right sensor detecting the line.

Using the formula for conditional probability, we can calculate the likelihood that the robot should turn right when the right sensor detects the line.

$$P(\text{Turn right} \mid \text{Right sensor detects line}) = P(\text{Turn right and Right sensor detects line}) / P(\text{Right sensor detects line})$$

Example Calculation

Suppose in tests:

- The probability that the robot turns right when the right sensor detects the line is 80% (0.80).
- The probability of the right sensor detecting the line is 50% (0.50).

$$P(\text{Turn right and Right sensor detects line}) = 0.8 \times 0.50 = 0.40$$

$$\text{Conditional probability: } P(\text{Turn right} \mid \text{Right sensor detects line}) = 0.4 / 0.5 = 0.8$$

This shows that when the right sensor detects the line, there is an 80% probability that the robot will turn right.

Fuzzy Logic and Probability

Fuzzy logic is used in situations where data from sensors are not definitive. For instance, sensors detecting a black line might not always be 100% certain. Fuzzy logic better manages these uncertainties.

Fuzzy Rules and Decision Making I

In fuzzy logic, data from sensors are processed according to specific rules. These rules, combined with probabilities, help determine the movement of the robot. Example Fuzzy Rules:

1. If the left sensor detects the line and the right sensor does not, then turn left.
2. If the right sensor detects the line and the left sensor does not, then turn right.
3. If both sensors detect the line, then go straight.
4. If neither sensor detects the line, then stop and recalibrate.

Example Application

1. Situation: The left sensor reports detecting the line with 70% certainty, while the right sensor reports 30%.
2. Fuzzy Logic Rules: In this case, since the left sensor detects the line with a higher probability, it is decided that the robot should turn left.

Conclusion

The concepts of probability and conditional probability can be used in the design of a line-following robot to evaluate information from sensors and adjust the robot's direction and speed. Furthermore, fuzzy logic allows for better management of uncertainties, enabling the robot to make more accurate decisions. These approaches offer valuable learning opportunities for STEM students, both theoretically and practically.

Autonomous Drone Design and Probability

The purpose of an autonomous drone is to follow a specified route safely and reach its destinations. During this process, various sensors on the drone detect obstacles and targets in the environment, guiding the drone's movement.

Application of Probability and Conditional Probability Sensor Data and Probabilities

Let's assume the drone is equipped with three obstacle detection sensors located at the front, left, and right. These sensors return a value of 1 when they detect an obstacle and 0 when they do not.

- **Sensor A (Front Sensor):** Returns 1 if it detects an obstacle ahead, otherwise 0.
- **Sensor B (Left Sensor):** Returns 1 if it detects an obstacle on the left, otherwise 0.
- **Sensor C (Right Sensor):** Returns 1 if it detects an obstacle on the right, otherwise 0.

Conditional Probability Calculations

Conditional probability can be used to decide which direction the drone should move based on the information from the sensors. Situation:

- **Event A:** The drone turning right (No obstacle on the right).
- **Event B:** The right sensor not detecting an obstacle.

Using the conditional probability formula, we can calculate that the drone should turn right when the right sensor does not detect an obstacle.

$$P(\text{Turn right} \mid \text{Right sensor did not detect an obstacle}) = P(\text{Turn right and Right sensor did not detect an obstacle}) / P(\text{Right sensor did not detect an obstacle})$$

Example Calculation

Suppose in tests:

- The probability of the drone turning right when the right sensor does not detect an obstacle is 70% (0.70).
- The probability of the right sensor not detecting an obstacle is 40% (0.40).

$$P(\text{Turn right and Right sensor did not detect an obstacle}) = 0.7 \times 0.40 = 0.28$$

Conditional probability:

$$P(\text{Turn right} \mid \text{Right sensor did not detect an obstacle}) = 0.28 / 0.40 = 0.70$$

This indicates that there is a 70% probability that the drone will turn right when the right sensor does not detect an obstacle.

Fuzzy Logic and Probability

Fuzzy logic is used when data from sensors are not definitive. For example, sensors may not always be 100% sure when detecting an obstacle. Fuzzy logic better manages these uncertainties.

Fuzzy Rules and Decision Making

In fuzzy logic, data from sensors are processed according to specific rules. These rules, along with probabilities, help determine the drone's movement. Example Fuzzy Rules:

1. If the front sensor detects an obstacle and the left sensor does not, then turn left.
2. If the front sensor detects an obstacle and the right sensor does not, then turn right.
3. If both side sensors do not detect an obstacle, then proceed straight.
4. If both side sensors detect an obstacle, then stop and recalibrate.

Example Application

1. Situation: The front sensor detects an obstacle, the left sensor reports a 60% probability of no obstacle, and the right sensor reports a 40% probability of no obstacle.
2. Fuzzy Logic Rules: In this case, since the left sensor reports a higher probability of no obstacle, it is decided that the drone should turn left.

Conclusion

The concepts of probability and conditional probability can be used in the design of an autonomous drone to evaluate information from sensors and adjust the drone's direction and speed. Additionally, fuzzy logic allows for better management of uncertainties, enabling the drone to make more accurate decisions. These approaches provide valuable learning opportunities for STEM students, both theoretically and practically.

Summary

- **Probability and Conditional Probability:** Allows calculating the drone's movement decisions based on sensor data.
- **Fuzzy Logic:** Evaluates sensor data more flexibly in uncertain situations and optimizes decision-making.
- **Example Scenarios:** Determines which direction the drone should move in based on sensor detection of obstacles.

With this information and examples, STEM students can learn to apply concepts of probability in autonomous drone design practically.

Independent and Dependent Events

The interaction between two or more events depends on whether the events are independent or dependent.

Independent Events

Independent events are those where the occurrence of one event does not affect the probability of the other event occurring.

Example:

Dice Rolling Consider rolling a fair die:

- **Event C:** The number shown is odd.
- **Event D:** The number shown is a multiple of 3.

Possible outcomes when rolling the die: 1, 2, 3, 4, 5, 6.

- **Event C (Odd Number):** {1, 3, 5}
- **Event D (Multiple of 3):** {3, 6}

Let's check if these events are independent:

- **Conditional Probability:** The probability of Event C occurring given that Event D has occurred:
 - $P(C|D) = P(C \cap D) / P(D)$
 - $P(C \cap D) = 1/6$
 - $P(D) = 2/6$
 - $P(C|D) = (1/6) / (2/6) = 1/6 \times 3 = 1/2$
 - $P(C) = 3/6 = 1/2$

As seen, $P(C|D) = P(C)$, which shows that Events C and D are independent.

Dependent Events

Dependent events are those where the occurrence of one event affects the probability of another event occurring.

Example: Card Drawing

Draw two cards randomly from a deck of 52 cards, without replacing the first before drawing the second.

- **Event A:** Drawing a king on the first draw.
- **Event B:** Drawing a king on the second draw after a king has been drawn first.

First Draw:

- There are 4 kings, and the total number of cards is 52. $P(A) = 4/52 = 1/13$

Second Draw:

- After the first draw, there are 51 cards left in the deck and 3 kings.

$$P(B|A) = 3/51 = 1/17$$

This shows that the events are dependent because the probability of the second event depends on the outcome of the first draw.

Summary

Independent Events:

- The occurrence of one event does not affect the probability of another event.
- For example, the events of rolling an odd number and a multiple of 3 when rolling a die are independent.

Dependent Events:

- The occurrence of one event affects the probability of another event.
- For example, drawing a king as the first card and then drawing a king as the second card from a deck are dependent events.

Application for STEM Students: Autonomous Drone

An autonomous drone can evaluate whether sensor data are independent or dependent while making decisions using the information from its sensors.

Example: Sensor Data and Independent/Dependent Events

- **Independent Events:**

- **Event A:** The front sensor detects an obstacle.
- **Event B:** The side sensor detects an obstacle.
- These events may be independent because the obstacle detected by the front sensor does not influence the detection of an obstacle by the side sensor.

- **Dependent Events:**

- **Event A:** The front sensor detects an obstacle.
- **Event B:** The drone turns to the right.
- These events are dependent because the detection of an obstacle by the front sensor increases the likelihood of the drone turning right.

Line-Following Robot and Events

A line-following robot uses data from sensors to follow a line drawn on the ground. This sensor data influences the robot's movement decisions.

Independent and Dependent Events

The characteristics of the interaction between two or more events depend on whether the events are independent or dependent.

Independent Events

Independent events are those where the occurrence of one event does not affect the probability of another event occurring.

- **Example: Sensors on a Line-Following Robot** Assume a line-following robot has three sensors: left, center, and right. These sensors return a value of 1 when they detect the line and 0 when they do not.
 - **Event A:** The left sensor detects the line.
 - **Event B:** The right sensor detects the line. Let's check if these events are independent:
 - Since the left and right sensors detect the line at different locations, these events might be independent. That is, the detection of the line by the left sensor does not affect the detection by the right sensor.

Example Calculation:

- Probability that the left sensor detects the line: $P(A) = 0.4$
- Probability that the right sensor detects the line: $P(B) = 0.3$

Conditional Probability:

The probability that the left sensor detects the line given that the right sensor has detected the line:

$$P(A|B) = P(A \cap B) / P(B)$$

- If $P(A \cap B) = P(A) \times P(B)$, the events are independent. $P(A \cap B) = 0.4 \times 0.3 = 0.12$

Conditional probability:

$P(A|B) = 0.12 / 0.3 = 0.4$ As seen, $P(A|B) = P(A)$, indicating that Events A and B are independent.

Dependent Events

Dependent events are those where the occurrence of one event affects the probability of another event occurring.

The Complement of an Event

The complement of an event (A^c) includes the scenarios where event A does not occur. In other words, the complement of an event consists of all outcomes in the sample space that are not part of the event.

Basic Concept

- **Event A:** A specific event.
- **Complement of Event A (A^c):** All situations where Event A does not occur.

The probability of the complementary event is equal to one minus the probability of the event:

$$P(A^c) = 1 - P(A)$$

Explanation with Examples

Example 1: Dice Rolling

Situation:

We roll a die and calculate the probability of getting a 6.

- **Event A:** The die shows a 6.
- **Complement of Event A (A^c):** The die shows a number other than 6.

$$P(A) = 1/6$$

$$A^c = \{1, 2, 3, 4, 5\} \quad P(A^c) = 1 - P(A) = 1 - 1/6 = 5/6$$

Explanation:

Whenever the die is rolled and a 6 does not come up (when it shows 1, 2, 3, 4, or 5), the complement of Event A occurs.

Example 2: Coin Tossing

Situation: We toss a coin and calculate the probability of getting heads.

- **Event A:** The coin shows heads.
- **Complement of Event A (A^c):** The coin shows tails.

$$P(A) = 1/2$$

$$A^c = \{\text{Tails}\} \quad P(A^c) = 1 - 1/2 = 1/2$$

Explanation:

Whenever the coin is tossed and heads does not come up (when it shows tails), the complement of Event A occurs.

Example 3: Line-Following Robot

Situation:

Consider a line-following robot that makes decisions based on data from its sensors. Let's calculate the probability that the left sensor detects the line.

- **Event A:** The left sensor detects the line.

$$P(A) = 0.7$$

- **Complement of Event A (A^c):** The left sensor does not detect the line.

$$A^c = \{\text{Left sensor does not detect the line}\}$$

$$P(A^c) = 1 - P(A) = 1 - 0.7 = 0.3$$

Explanation:

The complement of Event A occurs whenever the left sensor does not detect the line. In this case, the robot might make another movement, such as turning right or continuing straight.

Importance of the Complement of an Event

The complement of events is particularly important in probability calculations and decision-making processes. Knowing the probability of an event not occurring is as valuable as knowing the probability of it occurring.

Example 4: Classroom Test Results

Situation: Let's calculate the probability of a student passing a test in a classroom.

- **Event A:** A student passes the test.

$$P(A) = 0.8$$

- **Complement of Event A (A^c):** A student fails the test.

$$P(A^c) = 1 - P(A) = 1 - 0.8 = 0.2$$

Explanation:

The probability of a student failing the test is 20%.

Example 5: Weather Forecast

Situation: Let's calculate the probability of rain according to a weather forecast.

- **Event A:** It rains.

$$P(A) = 0.3$$

- **Complement of Event A (A^c):** It does not rain.

$$P(A^c) = 1 - P(A) = 1 - 0.3 = 0.7$$

Explanation:

The probability of it not raining is 70%.

Probability Addition Theorem

The probability addition theorem is used to calculate the likelihood of multiple events occurring together. This theorem is particularly important in cases where events either exclude each other or do not.

Basic Concepts

Mutually Exclusive Events

Mutually exclusive events are events that cannot occur at the same time. If events are mutually exclusive, the occurrence of one event prevents the occurrence of the other.

Non-Mutually Exclusive Events

Non-mutually exclusive events are events that can occur simultaneously. These types of events can occur together.

Addition Theorem

Mutually Exclusive Events

If $A_1, A_2, \dots, A_n$ are mutually exclusive events, the probability that at least one of these events occurs is the sum of their probabilities:

$$P(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n) = P(A_1) + P(A_2) + P(A_3) + \dots + P(A_n)$$

Non-Mutually Exclusive Events

If events A and B are not mutually exclusive (i.e., they can occur together), the probability of their union is calculated as follows:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Explanation with Examples

Example 1: Mutually Exclusive Events

Situation: We roll a die and calculate the probability of getting either an odd or an even number.

- **Event A:** Getting an odd number (1, 3, 5). $P(A) = 3/6 = 1/2$
- **Event B:** Getting an even number (2, 4, 6). $P(B) = 3/6 = 1/2$

The probability of getting either an odd or an even number:

$$P(A \cup B) = P(A) + P(B) = 1/2 + 1/2 = 1$$

Explanation:

The probability of getting either an odd or even number is 1 because every roll of the die will necessarily result in either an odd or an even number.

Example 2: Non-Mutually Exclusive Events

Situation: A newspaper vendor distributes two newspapers, referred to as X and Y, to homes in a large city.

- **Event A:** Homes that receive Newspaper X.
 $P(A) = 0.6$
- **Event B:** Homes that receive Newspaper Y.
 $P(B) = 0.45$
- **Event $A \cup B$:** Homes that receive both Newspapers X and Y.
 $P(A \cup B) = 0.8$

Given:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Substituting the values,

$$\text{we get: } 0.8 = 0.6 + 0.45 - P(A \cap B)$$

Solving this:

$$P(A \cap B) = 0.6 + 0.45 - 0.8 = 0.25$$

Result:

25% of the homes receive both newspapers.

Application for STEM Students: Line-Following Robot

Situation: Sensor Operation Let's assume a line-following robot has three sensors: left, center, and right sensors. These sensors return a value of 1 when they detect the line and 0 when they do not.

- **Event A:** The left sensor detects the line.
- **Event B:** The center sensor detects the line.
- **Event C:** The right sensor detects the line.

We assume that these events are not mutually exclusive and can occur simultaneously.

Probability Calculations

- **Event A:** The left sensor detects the line. $P(A) = 0.5$
- **Event B:** The center sensor detects the line. $P(B) = 0.7$
- **Event C:** The right sensor detects the line. $P(C) = 0.4$

Combined Probabilities The probability that either the left or the center sensor detects the line: $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.5 + 0.7 - 0.35 = 0.85$

The probability that either the center or the right sensor detects the line: $P(B \cup C) = P(B) + P(C) - P(B \cap C) = 0.7 + 0.4 - 0.28 = 0.82$

These examples help STEM students understand the probability addition theorem and how events are combined. With this knowledge, correct decisions can be made using sensor data from a line-following robot.

Product Probability Theorem

The product probability theorem is used to calculate the likelihood of two events occurring at the same time. This theorem is applied differently depending on whether the events are independent or dependent.

Independent Events

If two events A and B are independent, the probability of both events occurring together is equal to the product of their probabilities:

$$P(A \cap B) = P(A) \times P(B)$$

Dependent Events

If two events A and B are dependent, the probability of both A and B occurring together is equal to the probability of B multiplied by the probability of A given B:

$$P(A \cap B) = P(A \setminus B) \times P(B)$$

Explanation with Examples

Example 1: Independent Events

Situation:

Consider the events of rolling a die and flipping a coin.

- **Event A:** Rolling a 4 on the die.

$$P(A) = 1/6$$

- **Event B:** Coin landing on heads.

$$P(B) = 1/2 \quad P(B) = 1/2 \quad P(B) = 1/2$$

These two events are independent because the outcome of the die does not affect the outcome of the coin. Therefore, the probability of both events occurring simultaneously:

$$P(A \cap B) = P(A) \times P(B) = 1/6 \times 1/2 = 1/12$$

Example 2: Dependent Events

Situation:

Draw two cards randomly from a deck of 52 cards, without replacing the first before drawing the second.

- **Event A:** Drawing a king on the first draw.

$$P(A) = 4/52 = 1/13$$

- **Event B:** Drawing a king on the second draw, given that the first card was a king. After drawing a king first, there are 3 kings left and 51 cards in total.

$$P(B|A) = 3/51$$

These two events are dependent because the probability of the second card being a king depends on whether the first card was a king. The probability of both events occurring together:

$$P(A \cap B) = P(A) \times P(B|A) = 1/13 \times 3/51 = 3/663 = 1/221$$

Example: Electronic System Components

Situation:

An electronic system consists of four components, each with the same functionality. The functionality of each component is independent of the others, and the probability of each component being functional is 90%.

- **Event A1:**

Component 1 is functional.

$$P(A1)=0.9$$

- **Event A2:**

Component 2 is functional.

$$P(A2)=0.9$$

- **Event A3:**

Component 3 is functional.

$$P(A3)=0.9$$

- **Event A4:**

Component 4 is functional.

$$P(A4)=0.9$$

Since these components are independent, the probability of all components being functional:

$$P(A1 \cap A2 \cap A3 \cap A4) = P(A1) \times P(A2) \times P(A3) \times P(A4) = 0.9 \times 0.9 \times 0.9 \times 0.9 = 0.6561$$

Probability of the System Being Non-Functional

- **Event A1^c:**

Component 1 is non-functional.

$$P(A1^c) = 1 - P(A1) = 1 - 0.9 = 0.1$$

- **Event A2^c:**

Component 2 is non-functional.

$$P(A2^c) = 0.1$$

- **Event A3^c:** Component 3 is non-functional.

$$P(A3^c) = 0.1$$

- **Event A4^c:**

Component 4 is non-functional.

$$P(A4^c) = 0.1$$

The probability of all components being non-functional:

$$P(A1^c \cap A2^c \cap A3^c \cap A4^c) = 0.1 \times 0.1 \times 0.1 \times 0.1 = 0.0001$$

Probability of the System Being Functional

Probability of the system being non-functional:

$$P(\text{System non-functional}) = 0.0001$$

Probability of the system being functional:

$$P(\text{System functional}) = 1 - P(\text{System non-functional}) = 1 - 0.0001 = 0.9999$$

$$1 - P(\text{System non-functional}) = 1 - 0.0001 =$$

$$0.9999$$

This example demonstrates the application of probability concepts to calculate the likelihood of all electronic system components functioning together, highlighting the importance of understanding independence and dependence among system components in practical applications.

Application for STEM Students: Line-Following Robot Situation:

Sensor Operation

Let's assume a line-following robot has three sensors: left, center, and right sensors. These sensors return a value of 1 when they detect the line and 0 when they do not. Assume the sensors operate independently.

- **Event A1:** The left sensor detects the line.
 $P(A1)=0.7$
- **Event A2:** The center sensor detects the line.
 $P(A2)=0.8$
- **Event A3:** The right sensor detects the line.
 $P(A3)=0.6$

Since these events are independent, the probability of all sensors detecting the line is:

$$P(A1 \cap A2 \cap A3) = P(A1) \times P(A2) \times P(A3) = 0.7 \times 0.8 \times 0.6 = 0.336$$

These examples help STEM students understand the product probability theorem and how events are combined. With this knowledge, accurate decisions can be made using sensor data from a line-following robot.

What is Bayes' Theorem?

Bayes' Theorem allows for the updating of the probability of an event based on additional information about the occurrence of that event. This theorem helps calculate updated (posterior) probabilities using prior probabilities in light of new information.

Bayes' Theorem Formula

Bayes' Theorem for events A and B is expressed as:

$$P(A|B) = P(B|A) \times P(A) / P(B)$$

Where:

- **$P(A|B)$: Probability of A given B has occurred (posterior probability).**
- **$P(B|A)$: Probability of B given A has occurred.**
- **$P(A)$: Probability of A (prior probability).**
- **$P(B)$ Probability of B.**

Example: Success of a New Product in the Market

A company is ready to launch a new product and wants to determine its likelihood of success in the market. The company knows the probabilities of a product being a success, disaster, or neutral based on past data:

- **Success Probability ($P(A1)$): 60% (0.6)**
- **Disaster Probability ($P(A2)$): 10% (0.1)**
- **Neutral Scenario Probability ($P(A3)$): 30% (0.3)**

The company also knows the probability of receiving a positive outcome from a market research survey in each scenario:

- **Positive Outcome Probability in Success ($P(B|A1)$): 75% (0.75)**
- **Positive Outcome Probability in Disaster ($P(B|A2)$): 10% (0.1)**
- **Positive Outcome Probability in Neutral ($P(B|A3)$): 40% (0.4)**

The company wants to know the revised probabilities of each outcome when the market research survey shows a positive result.

Updating Probabilities Using Bayes' Theorem

Step 1: Calculate the Probability of B

The probability of B is calculated as the sum of the probabilities of all scenarios:

$$P(B) = P(B|A1) \times P(A1) + P(B|A2) \times P(A2) + P(B|A3) \times P(A3)$$

$$P(B) = (0.75 \times 0.6) + (0.1 \times 0.1) + (0.4 \times 0.3)$$

$$P(B)=0.45+0.01+0.12=0.58$$

Step 2: Calculate Updated Probabilities

- Updated Probability for Success

$$P(A1\backslash B)=P(B|A1)\times P(A1)/P(B)$$

$$P(A1\backslash B)=0.75\times 0.60/.58=0.450/.58\approx 0.776$$

- Updated Probability for Disaster

$$P(A2\backslash B)=P(B|A2)\times P(A2)/P(B)$$

$$P(A2\backslash B) = 0.01/0.58\approx 0.017$$

- Updated Probability for Neutral

$$P(A3\backslash B)=P(B|A3)\times P(A3)/P(B)$$

$$P(A3\backslash B)=0.4\times 0.3/0.58=0.120/.58\approx 0.207$$

Result When the market research survey shows a positive result:

- Probability of the product being successful: 77.6% (0.776)
- Probability of the product resulting in disaster: 1.7% (0.017)
- Probability of the product being neutral: 20.7% (0.207)

These results demonstrate how market research provides more information about the success of the product and guides the decision-making process.

Application for STEM Students: Line-Following Robot

Situation: Using data from sensors of a line-following robot, we can determine whether it is on the correct path by using Bayes' Theorem.

Probabilities:

- **Probability of Being on the Right Path ($P(A1)$):** %70 (0.7)
- **Probability of Being on the Wrong Path ($P(A2)$):** %30 (0.3)

Sensor Data:

- **Probability of a Positive Result when on the Right Path ($P(B|A1)$):** %90 (0.9)
- **Probability of a Positive Result when on the Wrong Path ($P(B|A2)$):** %20 (0.2)

Using Bayes' Theorem to Update Probabilities

Step 1: Calculate the Probability of B

$$P(B) = P(B|A1) \times P(A1) + P(B|A2) \times P(A2)$$

$$P(B) = (0.9 \times 0.7) + (0.2 \times 0.3)$$

$$P(B) = 0.63 + 0.06 = 0.69$$

Step 2: Calculate Updated Probabilities

Probability of Being on the Right Path ($P(A1|B)$):

$$P(A1|B) = P(B|A1) \times P(A1) / P(B)$$

$$P(A1|B) = 0.9 \times 0.7 / 0.69 = 0.63 / 0.69 \approx 0.913$$

Probability of Being on the Wrong Path ($P(A2|B)$):

$$P(A2|B) = P(B|A2) \times P(A2) / P(B)$$

$$P(A2|B) = 0.2 \times 0.3 / 0.69 = 0.06 / 0.69 \approx 0.087$$

Result

When sensors show a positive result:

- Probability of the robot being on the right path: 91.3% (0.913)
- Probability of the robot being on the wrong path: 8.7% (0.087)

These examples help STEM students understand Bayes' Theorem and the process of updating probabilities. With this knowledge, correct decisions can be made using sensor data from a line-following robot.

Scenario: Autonomous Drone and Bayes' Theorem

An autonomous drone uses sensors to make decisions while avoiding obstacles in a specific area. There are two possible states it can encounter:

1. **There is an obstacle (E).**
2. **There is no obstacle ($\neg E$).**

The drone evaluates these conditions based on the data from its sensors. The sensors can accurately detect obstacles:

- **Probability of detecting an obstacle when there is one ($P(S|E)$): %90 (0.9)**
- **Probability of not detecting an obstacle when there isn't one ($P(\neg S|\neg E)$): %80 (0.8)**

However, sensors can sometimes give false positives or false negatives:

- **False Positive Probability (detecting an obstacle when there isn't one) ($P(S|\neg E)$): %20 (0.2)**
- **False Negative Probability (not detecting an obstacle when there is one) ($P(\neg S|E)$): 10% (0.1)**

Scenario: When the Sensor Detects an Obstacle

Using Bayes' Theorem, we can update our belief about the presence of an obstacle when the sensor detects one (S):

$$P(E|S) = P(S|E) \times P(E) / P(S)$$

Step 1: Calculate the Probability of the Sensor Detecting an Obstacle ($P(S)$)

$$P(S) = P(S|E) \times P(E) + P(S|\neg E) \times P(\neg E)$$

$$P(S) = (0.9 \times 0.3) + (0.2 \times 0.7)$$

$$P(S) = 0.27 + 0.14 = 0.41$$

Step 2: Calculate the Updated Probability ($P(E|S)$)

$$P(E|S) = P(S|E) \times P(E) / P(S)$$

$$P(E|S) = 0.9 \times 0.3 / 0.41 = 0.27 / 0.41 \approx 0.659$$

Scenario: When the Sensor Does Not Detect an Obstacle

Using Bayes' Theorem, we can update our belief about the absence of an obstacle when the sensor does not detect one ($\neg S$):

$$P(\neg E|\neg S) = P(\neg S|\neg E) \times P(\neg E) / P(\neg S)$$

Step 1: Calculate the Probability of the Sensor Not Detecting an Obstacle ($P(\neg S)$)

$$P(\neg S) = P(\neg S|E) \times P(E) + P(\neg S|\neg E) \times P(\neg E)$$

$$P(\neg S) = (0.1 \times 0.3) + (0.8 \times 0.7)$$

$$P(\neg S) = 0.03 + 0.56 = 0.59$$

Step 2: Calculate the Updated Probability ($P(\neg E|\neg S)$)

$$P(\neg E|\neg S) = P(\neg S|\neg E) \times P(\neg E) / P(\neg S)$$

$$P(\neg E|\neg S) = 0.8 \times 0.7 / 0.59 = 0.560.59 \approx 0.949$$

Results

- When the sensor detects an obstacle: The real probability of there being an obstacle is 65.9% (0.659).
- When the sensor does not detect an obstacle: The real probability of there being no obstacle is 94.9% (0.949).

This example demonstrates how the drone can use sensor data to update its flight decisions. Bayes' Theorem helps the drone to assess its environmental situation more accurately based on sensor data, enabling safer and more efficient flights.

What is a Random Variable?

A random variable is a type of variable whose value is dependent on the numerical outcomes of certain random events. These variables must be measurable and typically take the form of real numbers. Random variables are used to quantify the outcomes of a random event.

Types of Random Variables

1. **Discrete Random Variable:**

Can take specific, countable values. For example, the outcome of rolling a die.

2. **Continuous Random Variable:**

Can take an infinite number of values within a certain range. For example, a person's height.

Example: Rolling Three Dice

Consider the outcome of rolling three dice as a random variable, XXX , representing the total result of the three dice.

Step 1: Number of Possible Outcomes

- Total number of dice: 3
- Each die can take a value between 1 and 6.
- Total number of outcomes: $6 \times 6 \times 6 = 216$
- Thus, there are 216 possible outcomes when three dice are rolled.

Step 2: Possible Values of the Random Variable

The random variable X represents the total of three dice. This total can range from 3 (when each die shows a 1) to 18 (when each die shows a 6).

$$X = \{3, 4, 5, \dots, 18\}$$

Step 3: Example Calculation

- Scenario: Calculate the probability that the total outcome is 10 when three dice are rolled.
- Possible Combinations: (4,3,3), (3,4,3), (3,3,4), (2,4,4), (4,2,4), (4,4,2), and so forth.
- Total Number of Combinations for a Sum of 10: Suppose there are 27 combinations that total 10.

The probability of the total being 10 is:

$$P(X=10) = \text{Number of combinations totaling 10} / \text{Total number of combinations} = 27/216 = 1/8$$

Real-Life Examples

Example 1: Grade Distribution in a Classroom

Consider the grades of students in a classroom as a random variable.

- **Random Variable (X):** The grades of the students.
- **Possible Values:** Any integer between 0 and 100.
- **Probability Distribution:** The likelihood of each grade can be calculated based on the distribution of past grades.

Example 2: Daily Temperature

The daily temperature in a city is a continuous random variable.

- **Random Variable (Y):** Daily temperature.
- **Possible Values:** Within a specific range (e.g., from -10°C to 40°C).
- **Probability Distribution:** The probability distribution of temperature values can be calculated based on historical weather data.

These examples help illustrate how random variables are used to model different types of data in real life, providing a basis for statistical analysis and decision making based on probabilistic outcomes.

Application for STEM Students: Line-Following Robot

Situation: Sensor Data

Let's consider the data from the sensors of a line-following robot as random variables.

- **Random Variable (S):** Values detected by the sensors.
- **Possible Values:** 0 (no line detected) or 1 (line detected).

Assume the robot has three sensors: Left, Center, Right.

Possible Combinations

- **Number of Combinations:** $2 \times 2 \times 2 = 8$ (Two possible values for each sensor).
- **Sample Combinations:**
 - (0,0,0): No sensor detects the line.
 - (1,0,0): Only the left sensor detects the line.
 - (1,1,1): All sensors detect the line.

Movement Decisions

- **Scenario (1,0,0):** The robot turns left.
- **Scenario (0,1,0):** The robot moves straight.
- **Scenario (0,0,1):** The robot turns right.

This example helps STEM students understand how random variables can be applied to real-world problems, such as robotics, to make decisions based on sensor data. By analyzing the possible states of the sensors, students can predict the robot's actions and refine its navigation algorithms.

Types of Random Variables

Random variables are categorized into two main types: discrete random variables and continuous random variables. These variables are used in probability theory to model different situations and have distinct characteristics.

Discrete Random Variable

Discrete random variables can only take a finite or countably infinite number of specific values. These values are typically integers and are selected from specified intervals.

Examples of Discrete Random Variables:

1. **Number of Typos on a Page**
 - The number of spelling errors on a page can be 0, 1, 2, 3, etc.
2. **Number of Customers in a Theater**
 - The number of customers in a theater at a given time can range from 0 up to the theater's capacity.
3. **Outcome of Rolling a Die**
 - The result of rolling a die can be 1, 2, 3, 4, 5, or 6.
4. **Number of Cars at a Gas Station**
 - The number of cars at a gas station at a certain time can be 0, 1, 2, 3, etc.

Continuous Random Variable

Continuous random variables can take any value within a certain interval. These variables can assume an infinite number of values and are typically expressed with a measurement unit.

Examples of Continuous Random Variables:

1. **Temperature**
 - The temperature in a room can be any value like 20.5°C, 21.0°C, 21.5°C, etc.
2. **Length**
 - The length of an object can be any value like 15.2 cm, 15.25 cm, 15.3 cm, etc.
3. **Time**
 - The duration of an event can be any value like 5.2 minutes, 5.25 minutes, 5.3 minutes, etc.
4. **Height**
 - A person's height can be any value like 160.5 cm, 161.0 cm, 161.5 cm, etc.

These examples help illustrate how discrete and continuous random variables are applied to real-world situations, offering a practical understanding for students and professionals in fields like statistics and engineering.

Explanation with Examples

Discrete Random Variable

Example 1: Number of Typos on a Page

A teacher counts the number of spelling errors on a student's written page. The number can be 0, 1, 2, 3, etc. Thus, the count of spelling errors is a discrete random variable.

Example 2: Outcome of Rolling a Die

When a die is rolled, the number that appears on its top face can be 1, 2, 3, 4, 5, or 6. These outcomes are finite and are integers, so the result of rolling a die is a discrete random variable.

Continuous Random Variable

Example 1: Temperature

The temperature recorded throughout a day in a city is a continuous random variable. The temperature can be any value, such as 20.5°C, 21.3°C, 22.8°C, etc.

Example 2: Time

The time it takes a runner to cover a certain distance is a continuous random variable. This duration can be any value, like 10.25 seconds, 10.5 seconds, 10.75 seconds, etc.

Application for STEM Students: Line-Following Robot

Discrete Random Variable: Sensor States

Consider a line-following robot equipped with three sensors: Left, Center, and Right. Whether these sensors detect the line can be expressed using discrete random variables.

- **Sensor State (X):** Whether the sensors detect the line.
 - **Possible Values:** 0 (not detected) or 1 (detected).

In this case, possible combinations for three sensors:

- (0,0,0)(0, 0, 0)(0,0,0): No sensors detect the line.
- (1,0,0)(1, 0, 0)(1,0,0): Only the left sensor detects the line.
- (1,1,1)(1, 1, 1)(1,1,1): All sensors detect the line.

Continuous Random Variable: Robot's Speed

Consider the robot's speed as a continuous random variable. We measure the robot's speed from 0 to 10 cm/s.

- **Speed (Y):** The speed of the robot.
 - **Possible Values:** 0 cm/s, 0.1 cm/s, 0.2 cm/s, ..., 10 cm/s.

The robot's speed can take any value within this range, making it a continuous random variable.

Summary

- **Discrete Random Variable:**

Can take a finite or countably infinite number of specific values. Examples include the outcome of rolling a die or the number of typos on a page.
- **Continuous Random Variable:**

Can take any value within a specified range. Examples include temperature or time.

This explanation helps clarify how discrete and continuous random variables are utilized in both theoretical and practical applications, enhancing understanding for students engaged in STEM education.

What is a Probability Distribution?

A probability distribution is a statistical function that describes all possible values and probabilities of a random variable within a specific range. It shows how the values of the random variable are distributed and the likelihood of each value occurring.

Elements of a Probability Distribution

- **Minimum and Maximum Values:** Define the boundaries of the distribution.
- **Mean:** Indicates the central tendency of the data.
- **Standard Deviation:** Measures the spread of the data.
- **Skewness:** Shows whether the distribution is symmetrical or not.
- **Kurtosis:** Indicates the peakedness or flatness of the distribution.

Visualization of Probability Distributions

Probability distributions can be visualized using the following methods:

- **Histogram:** Displays the distribution of data in the form of bars.
- **Density Plot:** Illustrates the density and probability density of the data.

Example: Perfect and Defective Parts in a Production Process

In a production process, both perfect and defective parts are produced. The following probability distribution shows the outcomes of this process:

Result	Probability
Perfect	0.85
Defective	0.15

This probability distribution indicates the likelihood of production outcomes. Approximately 85% of the parts are likely to be perfect (probability 0.85), while about 15% may be defective (probability 0.15).

Analysis of the Probability Distribution

Analyzing this probability distribution helps evaluate the distribution of defective and non-defective parts in production. This is crucial for quality control assessments, product acceptance decisions, and estimating rework or rejection rates.

- **Box Plot:** Shows the spread of the data and outliers.

Probability Distribution with Examples

Example 1: Rolling a Die

Situation: When rolling a die, the probability distribution of the number appearing on the top face.

- **Possible Values:** 1, 2, 3, 4, 5, 6
- **Probabilities:** The probability for each number is $1/6$

Value	Probability
1	$1/6$
2	$1/6$
3	$1/6$
4	$1/6$
5	$1/6$
6	$1/6$

This distribution shows that each number has an equal chance of appearing.

Example 2: Product Sales

Situation: The probability distribution of the number of products sold in a store in one day.

- **Possible Values:** 0, 1, 2, 3, 4, 5
- **Probabilities:**

Number of Sales	Probability
0	0.1
1	0.2
2	0.3
3	0.2
4	0.15
5	0.05

This distribution shows the probabilities of different numbers of products sold in one day.

Application for STEM Students: Autonomous Drone

Situation:

Sensor Data from a Drone

Let's examine the probability distribution of data from the sensors of an autonomous drone. The probabilities of the drone encountering an obstacle have been determined.

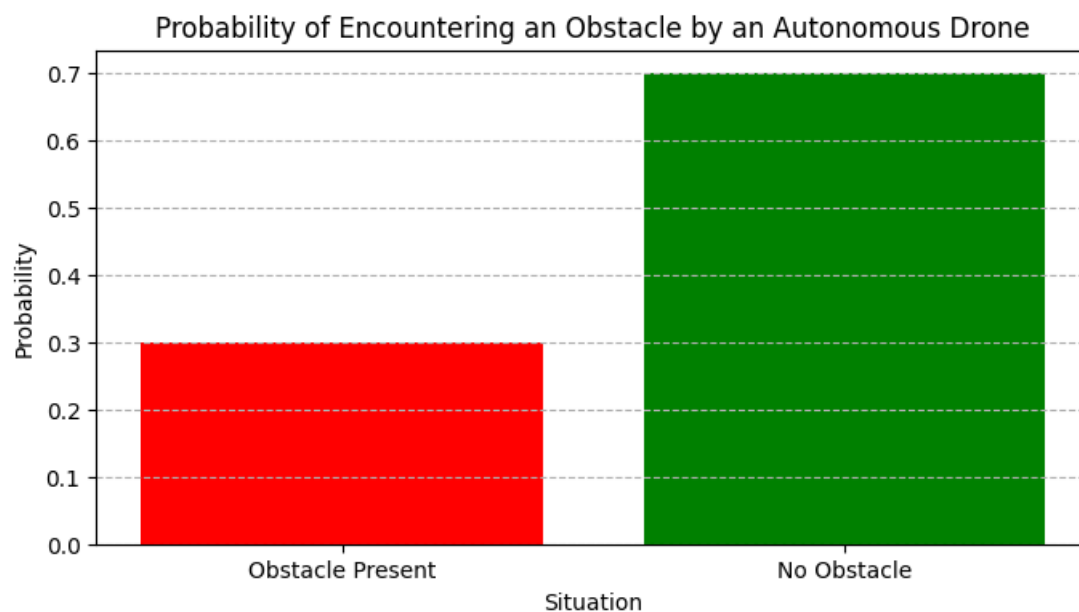
- **Possible Values:** Obstacle Present, No Obstacle
- **Probabilities:**

Situation	Probability
Obstacle Present	0.3
No Obstacle	0.7

This distribution shows the likelihood of the drone encountering an obstacle during its flight.

Visualization

We can visualize the probabilities of the drone encountering obstacles with a histogram:



```
import matplotlib.pyplot as plt
# Define the situations and their probabilities
situations = ['Obstacle Present', 'No Obstacle']
probabilities = [0.3, 0.7]

# Create a bar chart
```

```
plt.figure(figsize=(8, 4))
plt.bar(situations, probabilities, color=['red', 'green'])

# Adding title and labels
plt.title('Probability of Encountering an Obstacle by an Autonomous
Drone')
plt.xlabel('Situation')
plt.ylabel('Probability')

# Show the grid
plt.grid(axis='y', linestyle='--')

# Display the plot
plt.show()
```

Importance of Probability Distributions

Probability distributions help us understand how the values of random variables are distributed and the likelihood of these values occurring. This is crucial in decision-making processes, conducting risk analyses, and evaluating the probabilities of various events.

What is Expected Value?

Expected value is the weighted average of all possible outcomes of a random variable, weighted by their probabilities. In other words, the expected value represents the long-term average value that a random variable is expected to take.

Calculating Expected Value

The expected value is calculated based on the probability distribution of a random variable and is expressed by the following formula:

$$E[X] = \sum_{i=1}^n P(x_i) \times x_i$$

Where:

- $E[X]$ is the expected value.
- $P(x_i)$ is the probability of the value x_i
- x_i is the i -th value that the random variable can take.

Example: Expected Value of Rolling a Die

Let's calculate the expected value for the random variable X , which represents the outcome of rolling a die.

- **Possible Values:** 1, 2, 3, 4, 5, 6
- **Probability for each value:** $1/6$

$$E[X] = \frac{1}{6} \times (1 + 2 + 3 + 4 + 5 + 6)$$

$$E[X] = \frac{1}{6} \times 21$$

$$E[X] = 3.5$$

This shows that the average value observed over many rolls of the die is 3.5.

Probability Distribution and Expected Value

The expected value is calculated using the probability distribution and represents the theoretical average outcome that a random variable is expected to take over the long term.

Other Examples

Example 1: Number of Customers in a Theater

The probability distribution of the number of customers in a theater on a given day is as follows:

Number of Customers	Probability
0	0.1
1	0.2
2	0.3
3	0.2
4	0.15
5	0.05

Let's calculate the expected number of customers:

$$E[X] = (0 \times 0.1) + (1 \times 0.2) + (2 \times 0.3) + (3 \times 0.2) + (4 \times 0.15) + (5 \times 0.05)$$

$$E[X] = 0 + 0.2 + 0.6 + 0.6 + 0.6 + 0.25$$

$$E[X] = 2.25$$

This shows that the expected average number of customers in the theater on a given day is 2.25.

Example 2: Defective Parts in a Production Process

The probability distribution of the number of defective parts in a production process is as follows:

Number of Defective Parts	Probability
0	0.85
1	0.10
2	0.05

Let's calculate the expected number of defective parts:

$$E[X] = (0 \times 0.85) + (1 \times 0.10) + (2 \times 0.05)$$

$$E[X] = 0 + 0.10 + 0.10$$

$$E[X] = 0.20$$

This shows that the expected number of defective parts per batch in the production process is 0.20.

Application for STEM Students: Autonomous Drone

Situation: Drone Obstacle Detection

An autonomous drone uses sensors to detect obstacles. The probability distribution for obstacle detection is as follows:

Situation	Probability
Obstacle Present	0.3
No Obstacle	0.7

In the drone's obstacle detection process, we can calculate the expected value to determine the average number of obstacles detected at a given time.

Number of Obstacles	Probability
0	0.7
1	0.3

Let's calculate the expected number of obstacles:

$$E[X] = (0 \times 0.7) + (1 \times 0.3)$$

$$E[X] = 0 + 0.3$$

$$E[X] = 0.3$$

This shows that, on average, the drone is expected to detect 0.3 obstacles per flight.

Conclusion

The expected value is the average value calculated by weighting all possible outcomes of a random variable by their probabilities. This concept is used to understand probability distributions and to estimate long-term average outcomes. These examples help STEM students better understand the concept of expected value and its application in real-world scenarios.

Random Variable and Probability Distribution

Random Variable:

A variable that depends on the numerical outcomes of specific random events. These variables numerically represent the outcomes of an experiment. For example, the number shown on the top face when a die is rolled is a random variable.

Probability Distribution:

Defines all possible values that a random variable can take and the probabilities of these values. It shows how the different outcomes of a random variable are distributed and how likely each outcome is.

Application: Mechanical Hand Tool Experiment

Context:

A STEM team is conducting an experiment to test a newly developed mechanical hand tool. They are aiming to assess whether the tool works successfully by having 1,000 people use it.

Objective: The goal of the experiment is to report on the tool's failure rate and use this information in the product development process.

Scenario and Parameters:

- **Experiment:** 1,000 individuals test the mechanical hand tool.
- **Success:** The tool functions without issues.
- **Failure:** The tool malfunctions or causes problems.
- **Probability of Success (p):** Based on previous tests and estimates, the probability of success is 95% ($p = 0.95$).
- **Probability of Failure (1-p):** 5% (0.05)

This scenario can be modeled using a binomial distribution because each test can be considered an independent Bernoulli trial with two outcomes (success or failure).

Probability Calculations for Failure Count using Binomial Distribution:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

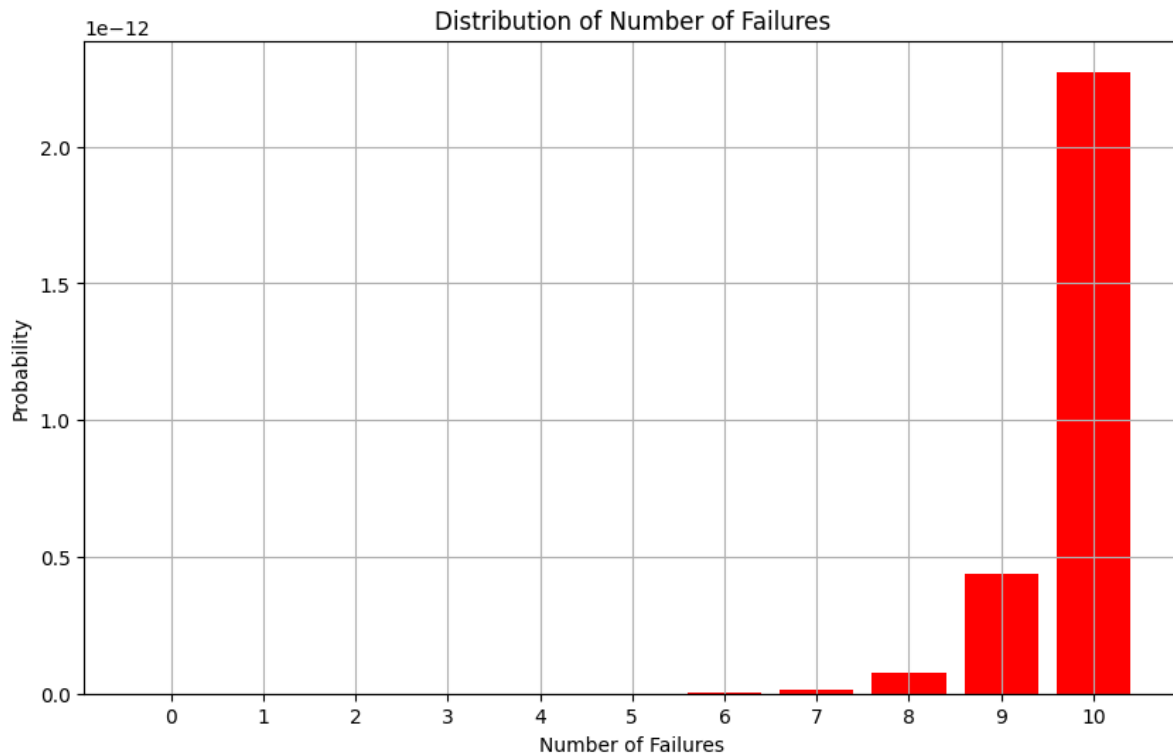
- **n:** Total number of trials (1,000)
- **k:** Number of failures
- **p:** Probability of success (95%)

Example Calculations:

- **Probability of no failures:** $P(X=0)$
- **Probability of exactly one failure:** $P(X=1)$

- **Probability of exactly two failures:** $P(X=2)$

Visualization: Plotting a Histogram We can provide students with an example of how to plot a histogram to visualize the results of the experiment. This will show the distribution of the number of failures.



```
import matplotlib.pyplot as plt
from scipy.stats import binom

# Experiment parameters
n = 1000
p = 0.95

# Possible number of failures
k_values = list(range(11)) # Displaying failure numbers from 0 to 10
probabilities = [binom.pmf(k, n, 1-p) for k in k_values]

# Plotting the histogram
plt.figure(figsize=(10, 6))
plt.bar(k_values, probabilities, color='red')
plt.title('Distribution of Number of Failures')
```

```
plt.xlabel('Number of Failures')  
plt.ylabel('Probability')  
plt.xticks(k_values)  
plt.grid(True)  
plt.show()
```

Reporting: Students can report the results of the experiment using this histogram. The histogram clearly displays the probability distribution of the number of failures, offering a clear understanding of the product's field performance. The report can include analysis, evaluations, and recommendations, supported by this visual representation.

This exercise not only helps in assessing the mechanical tool's reliability but also enhances students' understanding of statistical analysis and data visualization techniques in practical scenarios.

Discrete Probability Distribution

A discrete probability distribution describes scenarios where a random variable can take on distinct, countable values. These distributions help us calculate the probabilities of specific events.

Bernoulli Distribution

The Bernoulli distribution is used for experiments with exactly two possible outcomes: success (1) or failure (0). The key characteristic of this distribution is the probability of success p and the probability of failure $1-p$.

Example: Coin Toss

- **Experiment:** Tossing a coin.
- **Success:** Getting heads (1).
- **Failure:** Getting tails (0).
- **Probability of success (p):** 0.5
- **Probability of failure ($1-p$):** 0.5

The probability function is:

$$P(X = x) = p^x(1 - p)^{1-x}$$

where x can be 0 or 1.

Binomial Distribution

The binomial distribution calculates the number of successes in a fixed number of independent Bernoulli trials. The parameters of this distribution are the number of trials n and the probability of success p in each trial.

Example: Light Bulb Manufacturing Process

- **Experiment:** Producing 1000 light bulbs.
- **Success:** A bulb being defective.
- **Probability of success (p):** 0.05
- **Number of trials (n):** 1000

The probability function is:

$$P(X = k) = \binom{1000}{k} (0.05)^k (0.95)^{998-k}$$

where k is the number of defective bulbs.

For example, the probability of having 2 defective bulbs is:

$$P(X = 2) = \binom{1000}{2} (0.05)^2 (0.95)^{998}$$

Poisson Distribution

The Poisson distribution calculates the probability of a given number of events happening in a fixed interval of time or space. This distribution is used when events occur independently and at a constant average rate.

Example: Customer Arrival Rate

- **Experiment:** Number of customers arriving at a store per hour.
- **Average number of customers (λ):** 5

The probability function is:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

where k is the number of customers arriving in an hour.

For example, the probability of 3 customers arriving in one hour is:

$$P(X = 3) = \frac{5^3 e^{-5}}{3!}$$

Summary

- **Bernoulli Distribution:** Single trial with two outcomes (success/failure).

Example: Coin toss.

- **Binomial Distribution:** Fixed number of independent Bernoulli trials.

Example: Number of defective bulbs in a batch of 1000.

- **Poisson Distribution:** Number of events in a fixed interval.

Example: Number of customers arriving at a store per hour.

These discrete probability distributions help us understand and predict the outcomes of random events. For STEM students, these concepts are essential tools in probability and statistics.

Continuous Probability Distribution

A continuous probability distribution describes scenarios where a random variable can take on an infinite number of values within a certain range. Continuous distributions define probabilities over an interval rather than at specific points.

Normal Distribution

The normal distribution is one of the most widely used continuous distributions in probability theory. It is often referred to as the "bell curve" because its graph is bell-shaped.

- **Mean (μ):** The central point of the distribution.
- **Standard Deviation (σ):** Indicates the spread of the distribution.

Example: Student Grades

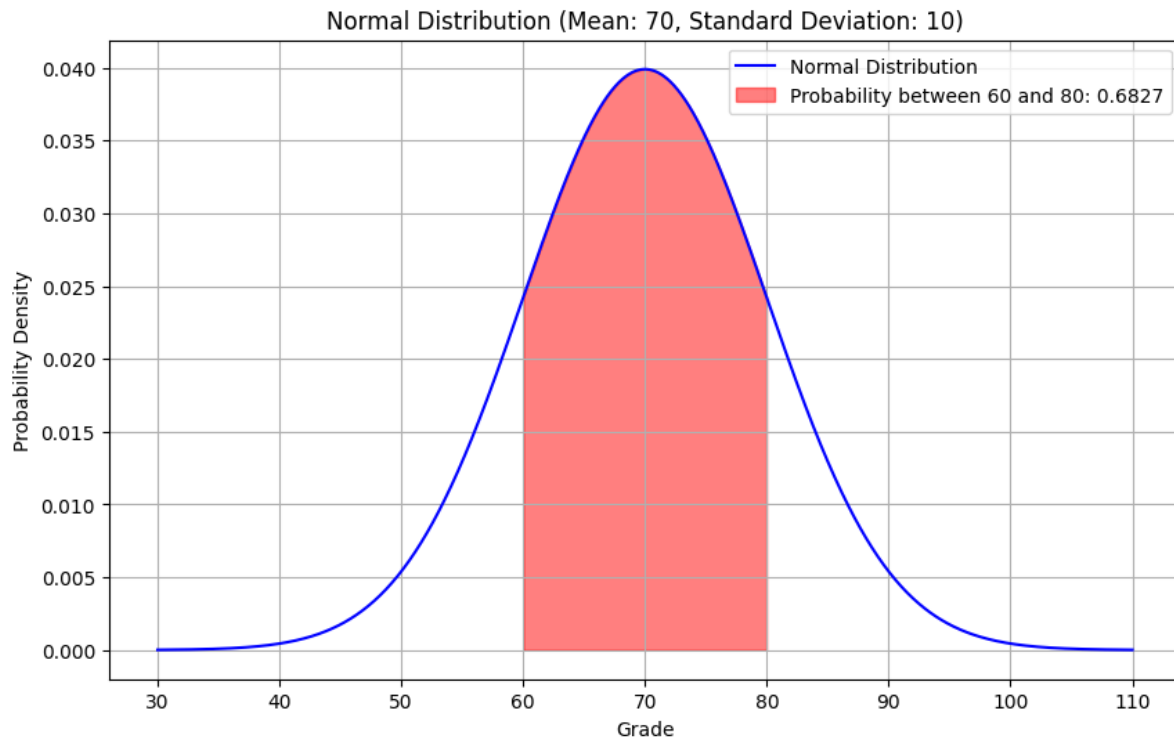
Assume that the grades of students in a class follow a normal distribution. Let's say the average grade is 70, and the standard deviation is 10.

By using the properties and equations of the normal distribution, we can determine various probabilities and insights about the distribution of students' grades.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

This formula is used to calculate the probabilities within a specific range of grades. For example, we can calculate the probability of getting a grade between 60 and 80.

Graphically, the normal distribution is shown as follows:



```
import numpy as np
import matplotlib.pyplot as plt
import scipy.stats as stats

# Normal distribution parameters
mu = 70    # Mean
sigma = 10 # Standard deviation

# Probability density function (PDF) of the normal distribution
def normal_pdf(x, mu, sigma):
    return (1 / (sigma * np.sqrt(2 * np.pi))) * np.exp(-0.5 * ((x - mu)
/ sigma) ** 2)

# Define x values
x = np.linspace(mu - 4 * sigma, mu + 4 * sigma, 1000)
y = normal_pdf(x, mu, sigma)

# Calculate probability for the range 60 to 80
prob_60_80 = stats.norm.cdf(80, mu, sigma) - stats.norm.cdf(60, mu,
sigma)

# Plot the graph
plt.figure(figsize=(10, 6))
```

```
plt.plot(x, y, label='Normal Distribution', color='blue')
plt.fill_between(x, 0, y, where=(x >= 60) & (x <= 80), color='red',
alpha=0.5, label=f'Probability between 60 and 80: {prob_60_80:.4f}')
plt.title('Normal Distribution (Mean: 70, Standard Deviation: 10)')
plt.xlabel('Grade')
plt.ylabel('Probability Density')
plt.legend()
plt.grid(True)
plt.show()
```

Uniform Distribution

Uniform distribution describes a scenario where a continuous random variable has an equal probability of taking any value within a specified range. This means every value within the range is equally likely.

- **Lower Bound (a):** The starting point of the distribution.
- **Upper Bound (b):** The endpoint of the distribution.

Example: Random Number Generation

Assume a computer program generates a random number between 0 and 10. This number follows a uniform distribution.

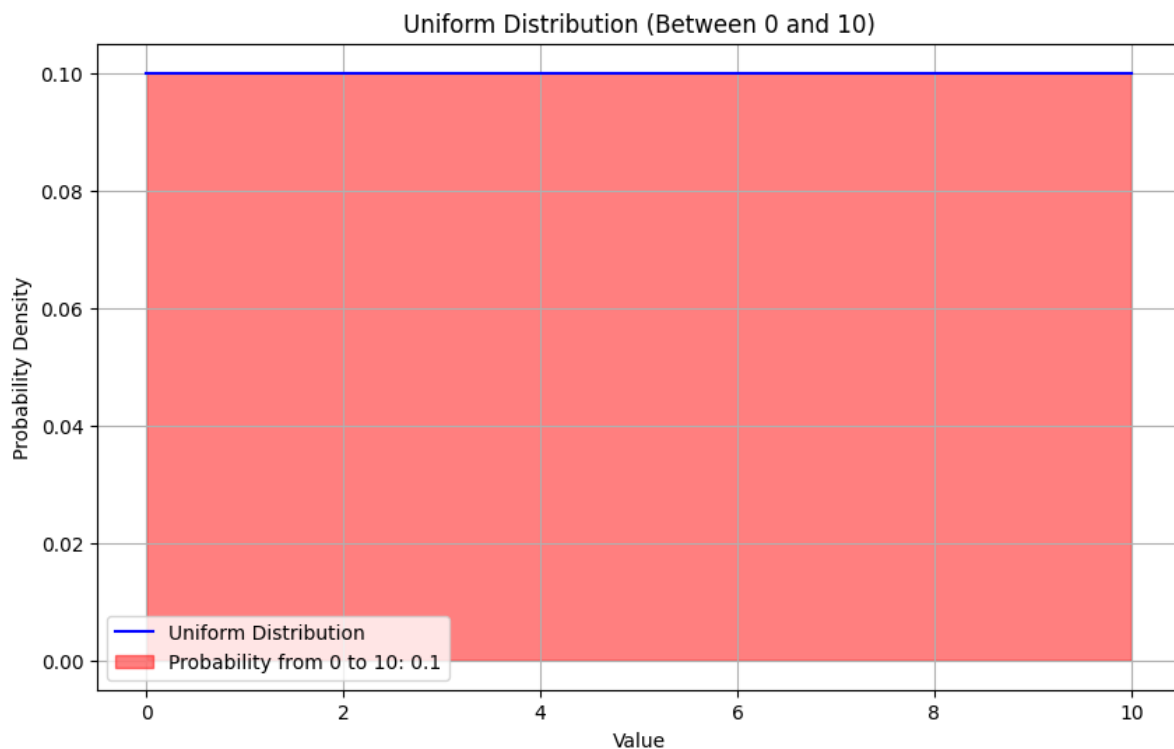
The probability density function (PDF) for a uniform distribution is defined as:

$$P(a \leq X \leq b) = \frac{1}{b-a}$$

The probability of any value between 0 and 10 is:

$$P(0 \leq X \leq 10) = \frac{1}{10-0} = 0.1$$

This shows that any number in this range has an equal probability of being chosen. Graphically, the uniform distribution is depicted as follows:



```
import numpy as np
import matplotlib.pyplot as plt

# Uniform distribution parameters
a = 0
b = 10

# Define x values
x = np.linspace(a, b, 1000)
y = np.full_like(x, 1 / (b - a))

# Plot the graph
plt.figure(figsize=(10, 6))
plt.plot(x, y, label='Uniform Distribution', color='blue')
plt.fill_between(x, 0, y, color='red', alpha=0.5, label='Probability from 0
to 10: 0.1')
plt.title('Uniform Distribution (Between 0 and 10)')
plt.xlabel('Value')
plt.ylabel('Probability Density')
plt.legend()
plt.grid(True)
plt.show()
```

Summary

- **Normal Distribution:** Symmetrical bell curve centered around the mean, commonly used for natural phenomena and human behaviors.

Example: Student grades, heights.

- **Uniform Distribution:** Flat distribution where all values within a specified range are equally likely.

Example: Random number generation in a computer program.

Understanding these continuous probability distributions is fundamental for modeling various natural and man-made processes. For STEM students, grasping these distributions is a cornerstone of statistics and probability theory.

Normal Probability Distribution

The normal distribution is a continuous probability distribution used to describe many natural phenomena. This distribution is characterized by a bell-shaped curve that is symmetrical around its mean.

Parameters of the Normal Distribution

The normal distribution is defined by two main parameters:

1. **Mean (μ):** Represents the center of the distribution and determines where the bell curve is centered.
2. **Standard Deviation (σ):** Measures the spread or dispersion of the distribution and determines the shape and width of the bell curve.

Properties of the Normal Distribution

- The mean, median, and mode are all the same.
- The distribution is symmetrical: half of the values lie below the mean, and half lie above the mean.
- The mean and standard deviation are the two main parameters that define the distribution.

Normal Distribution Equation

The normal distribution equation is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Where:

- x represents a specific value within the distribution.
- μ is the mean of the distribution.
- σ is the standard deviation.
- π is the constant Pi (approximately 3.14159).
- e is Euler's number (approximately 2.71828).

Probability Density Function (PDF)

The Probability Density Function (PDF) is a mathematical function that describes the probability distribution of a continuous random variable. The PDF has two main properties:

1. **Non-negativity:** The PDF is non-negative for all possible values of x .
2. **Total area equals 1:** The integral of the PDF over the entire range is equal to 1. This ensures that the total probability distribution is normalized to 1.

Example: Normal Probability Distribution

The normal distribution is a continuous probability distribution used to describe many natural phenomena. This distribution is characterized by a bell-shaped curve that is symmetrical around its mean.

A Fun Example: Student Heights

Let's assume the heights of students in a classroom are normally distributed. We'll use this example to explain the basic properties and calculations of the normal distribution.

Parameters of the Normal Distribution

1. **Mean (μ):** Represents the center of the distribution.
2. **Standard Deviation (σ):** Measures the spread or dispersion of the distribution.

In this example:

- Mean height (μ) = 160 cm
- Standard deviation (σ) = 10 cm

Properties of the Normal Distribution

- The mean, median, and mode are all the same.
- The distribution is symmetrical: half of the values lie below the mean, and half lie above the mean.
- The mean and standard deviation are the two main parameters that define the distribution.

Normal Distribution Equation

The normal distribution equation is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Where:

- x is the variable representing a value within the distribution.
- μ is the mean of the distribution.
- σ is the standard deviation.
- π is the constant Pi (approximately 3.14159).
- e is Euler's number (approximately 2.71828).

Example: Student Heights

Let's assume the heights of students in a classroom are normally distributed. The average height is 160 cm with a standard deviation of 10 cm. Our goal is to determine the probability that the students' heights are between 150 cm and 170 cm.

Step 1: Standardize the Values

Standardize the values of 150 and 170 using the formula:

$$z = \frac{x - \mu}{\sigma}$$

For 150 cm:

$$z_1 = \frac{150 - 160}{10} = -1$$

For 170 cm:

$$z_2 = \frac{170 - 160}{10} = 1$$

Step 2: Calculate Cumulative Probabilities

Use the standard normal distribution table to find the cumulative probabilities corresponding to the standardized values:

- For $z = -1$, the cumulative probability $P(Z \leq -1) \approx 0.1587$
- For $z = 1$, the cumulative probability $P(Z \leq 1) \approx 0.8413$

Step 3: Calculate the Desired Probability

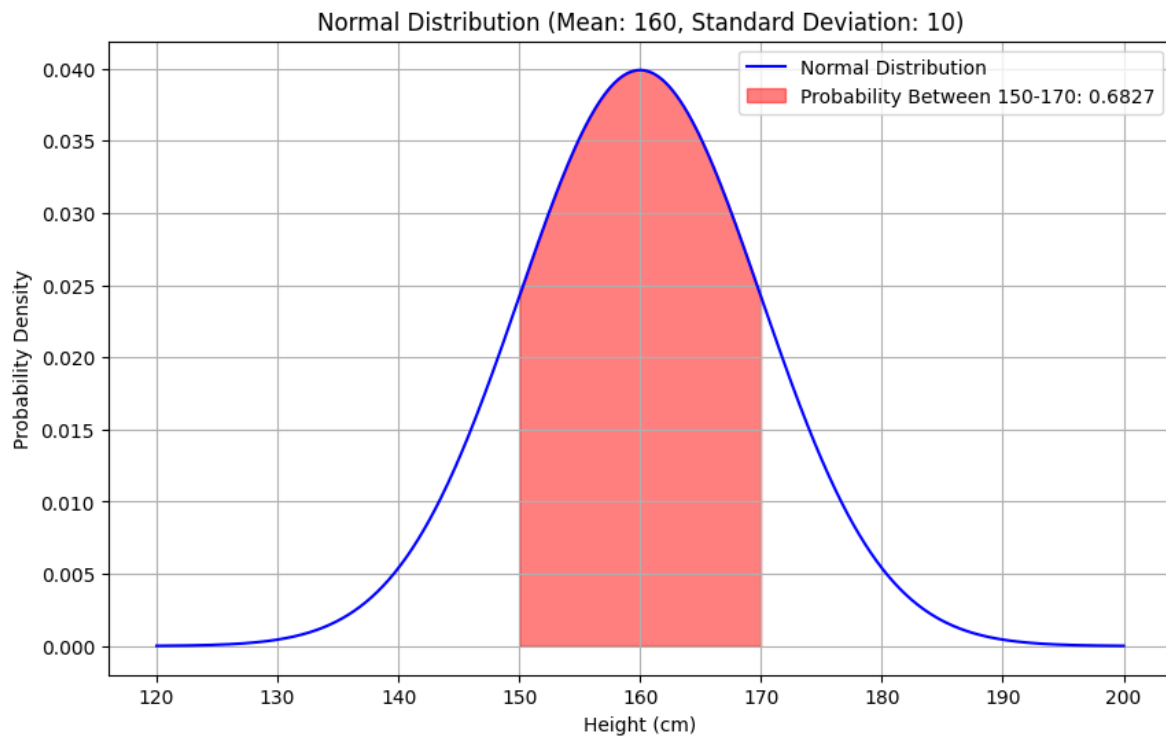
Find the probability between these two values:

$$P(150 \leq x \leq 170) = P(Z \leq 1) - P(Z \leq -1)$$

$$\approx 0.8413 - 0.1587$$

$$\approx 0.6826$$

This indicates that the probability of the students' heights being between 150 cm and 170 cm is approximately 68.26%.



```
import numpy as np
import matplotlib.pyplot as plt
import scipy.stats as stats

# Normal distribution parameters
mu = 160    # Mean height (cm)
sigma = 10  # Standard deviation (cm)

# Probability density function (PDF) of normal distribution
def normal_pdf(x, mu, sigma):
    return (1 / (sigma * np.sqrt(2 * np.pi))) * np.exp(-0.5 * ((x - mu) /
sigma) ** 2)

# Define x values
x = np.linspace(mu - 4 * sigma, mu + 4 * sigma, 1000)
y = normal_pdf(x, mu, sigma)

# Calculate probability between 150 and 170
prob_150_170 = stats.norm.cdf(170, mu, sigma) - stats.norm.cdf(150, mu,
sigma)
```

```
# Plotting the graph
plt.figure(figsize=(10, 6))
plt.plot(x, y, label='Normal Distribution', color='blue')
plt.fill_between(x, 0, y, where=(x >= 150) & (x <= 170), color='red',
alpha=0.5, label=f'Probability Between 150-170: {prob_150_170:.4f}')
plt.title('Normal Distribution (Mean: 160, Standard Deviation: 10)')
plt.xlabel('Height (cm)')
plt.ylabel('Probability Density')
plt.legend()
plt.grid(True)
plt.show()
```

Conclusion

The normal distribution is widely used to model many natural phenomena. It is a continuous probability distribution characterized by a bell-shaped curve that is symmetrical around its mean. Defined by its mean (μ) and standard deviation (σ), it allows for the calculation of probabilities within specific ranges using the PDF.

In this example, the probability of the students' heights being between 150 cm and 170 cm is approximately 68.26%.

Example: Balloon Height Competition

A school is organizing a balloon competition. The balloons' average height is 500 meters with a standard deviation of 50 meters. Let's assume the balloon heights follow a normal distribution.

Objective

Determine the probability that the balloons' heights are between 450 meters and 550 meters.

Step 1: Standardize the Values

Standardize the values of 450 and 550 using the formula:

$$z = \frac{x - \mu}{\sigma}$$

For 450 meters:

$$z_1 = \frac{450 - 500}{50} = -1$$

For 550 meters:

$$z_2 = \frac{550 - 500}{50} = 1$$

Step 2: Calculate Cumulative Probabilities

Use the standard normal distribution table to find the cumulative probabilities corresponding to the standardized values:

- For $z = -1$, the cumulative probability $P(Z \leq -1) \approx 0.1587$
- For $z = 1$, the cumulative probability $P(Z \leq 1) \approx 0.8413$

Step 3: Calculate the Desired Probability

Find the probability between these two values:

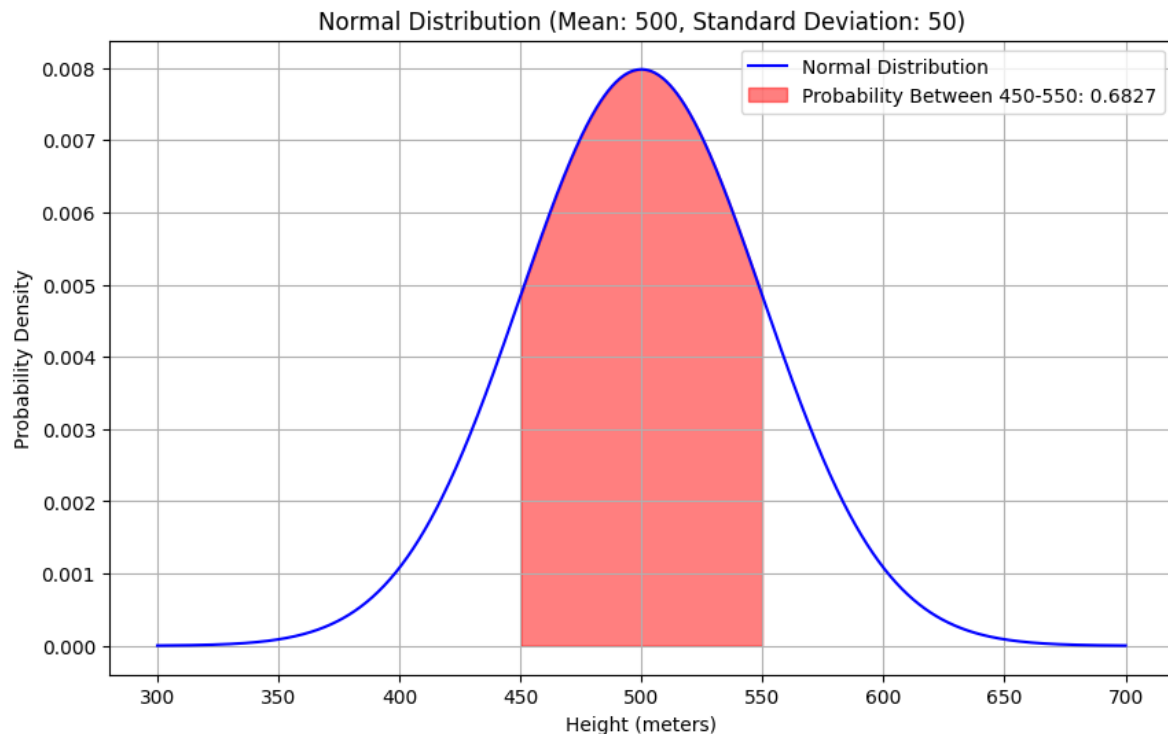
$$P(450 \leq x \leq 550) = P(Z \leq 1) - P(Z \leq -1)$$

$$\approx 0.8413 - 0.1587$$

$$\approx 0.6826$$

This indicates that the probability of the balloons being between 450 meters and 550 meters is approximately 68.26%.

Let's visualize this with a graph:



```
import numpy as np
import matplotlib.pyplot as plt
import scipy.stats as stats

# Normal distribution parameters
mu = 500    # Mean
sigma = 50  # Standard deviation

# Probability density function (PDF) of normal distribution
def normal_pdf(x, mu, sigma):
    return (1 / (sigma * np.sqrt(2 * np.pi))) * np.exp(-0.5 * ((x - mu)
/ sigma) ** 2)

# Define x values
x = np.linspace(mu - 4 * sigma, mu + 4 * sigma, 1000)
y = normal_pdf(x, mu, sigma)

# Calculate probability between 450 and 550
prob_450_550 = stats.norm.cdf(550, mu, sigma) - stats.norm.cdf(450, mu,
sigma)
```

```
# Plotting the graph
plt.figure(figsize=(10, 6))
plt.plot(x, y, label='Normal Distribution', color='blue')
plt.fill_between(x, 0, y, where=(x >= 450) & (x <= 550), color='red',
alpha=0.5, label=f'Probability Between 450-550: {prob_450_550:.4f}')
plt.title('Normal Distribution (Mean: 500, Standard Deviation: 50)')
plt.xlabel('Height (meters)')
plt.ylabel('Probability Density')
plt.legend()
plt.grid(True)
plt.show()
```

Result

The normal distribution is widely used to model many natural phenomena. It is a continuous probability distribution characterized by a bell-shaped curve that is symmetrical around its mean. Defined by its mean (μ) and standard deviation (σ), it allows for the calculation of probabilities within specific ranges using the PDF.

In this example, the probability of the balloons' heights being between 450 meters and 550 meters is approximately 68.26%.

Uniform Probability Distribution

A uniform probability distribution is a type of probability distribution in which all values within a certain range have an equal chance of occurring. This means that every value in the range appears with the same probability.

Parameters of Uniform Distribution

The uniform distribution is defined by two parameters:

1. **Lower Bound (a):** Represents the smallest value where the distribution starts.
2. **Upper Bound (b):** Represents the largest value where the distribution ends.

Characteristics of Uniform Distribution

- **Probability Density Function (PDF):** It is constant throughout the interval. This means that the probability of each value is equal.
- **Specific Interval:** It is defined over the interval $[a, b]$. Values outside this interval have a probability of zero.

Uniform Probability Distribution Equation

For a continuous random variable X defined over the interval $[a, b]$, the probability density function (PDF) of a continuous uniform distribution is given by:

$$f(x) = \frac{1}{b-a} \quad \text{for } a \leq x \leq b$$

$$f(x) = 0 \quad \text{otherwise}$$

Where:

- a is the lower bound of the interval.
- b is the upper bound of the interval.
- $b-a$ is the width of the interval.

Uniform Probability Distribution: Examples

Example 1: Drawing a Card from a Deck

Consider the event of drawing a card from a deck of playing cards. Each card has an equal chance of being selected. Therefore, the probability of drawing a heart, diamond, spade, or club is equal, with each having a probability of $1/4$.

Example 2: Coin Tossing

When you toss a coin, the probability of getting heads or tails is equal. Therefore, the probability of getting heads or tails is $1/2$ each.

Example 3: Waiting Time for a Bus

Consider a person waiting for a bus that arrives every 15 minutes. The exact arrival time of the bus varies randomly within this 15-minute window. This means the probability of the bus arriving at any particular minute within the 15 minutes is the same.

Parameters:

- $A=0$ minutes

- B=15 minutes

Probability Density Function (PDF):

$$f(x) = \frac{1}{15-0} = \frac{1}{15} \quad \text{for } 0 \leq x \leq 15$$

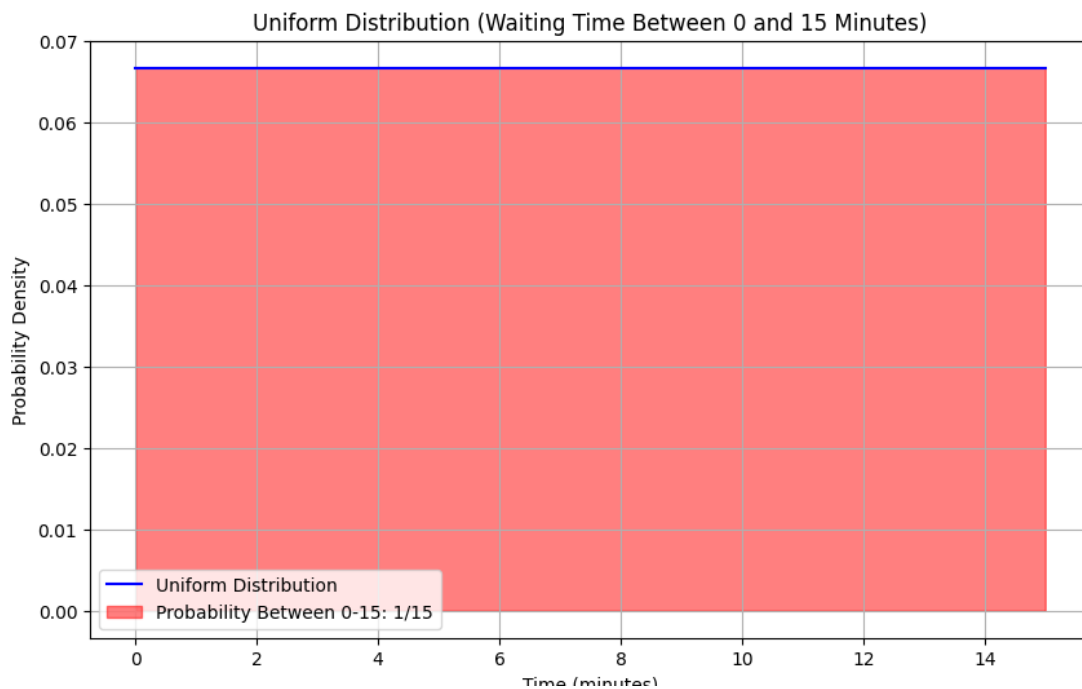
$$f(x) = 0 \quad \text{otherwise}$$

In this case, the probability of the bus arriving at any specific minute within the 15-minute interval is 1/15, and it is 0 for any time outside this interval.

Graphical Representation

A uniform distribution graph shows constant probability density over the interval [a,b]. The line is horizontal and straight across the interval.

Here is a simple example of a uniform distribution graph for bus waiting times between 0 and 15 minutes:



This graph shows that the probability of the bus arriving within 0 to 15 minutes is equal for each minute.

```
import numpy as np
import matplotlib.pyplot as plt

# Uniform distribution parameters
a = 0
b = 15

# Define x values
x = np.linspace(a, b, 1000)
y = np.full_like(x, 1 / (b - a))
```

```
# Plotting the graph
plt.figure(figsize=(10, 6))
plt.plot(x, y, label='Uniform Distribution', color='blue')
plt.fill_between(x, 0, y, color='red', alpha=0.5, label='Probability
Between 0-15: 1/15')
plt.title('Uniform Distribution (Waiting Time Between 0 and 15
Minutes)')
plt.xlabel('Time (minutes)')
plt.ylabel('Probability Density')
plt.legend()
plt.grid(True)
plt.show()
```

Conclusion

A uniform probability distribution describes situations where any value within a certain range is equally likely to be chosen. This distribution is used to model real-world scenarios like drawing a card from a deck, tossing a coin, and waiting times for a bus. Understanding these concepts helps STEM students grasp the fundamental principles of probability and statistics.

Example: Random Number Generation in Games

Imagine you are designing a computer game. In the game, the movement speed of characters is randomly determined between 5 m/s and 10 m/s. In this case, we can assume that the movement speeds follow a uniform distribution.

Parameters:

- $a=5a = 5a=5$ m/s
- $b=10b = 10b=10$ m/s

Probability Density Function (PDF):

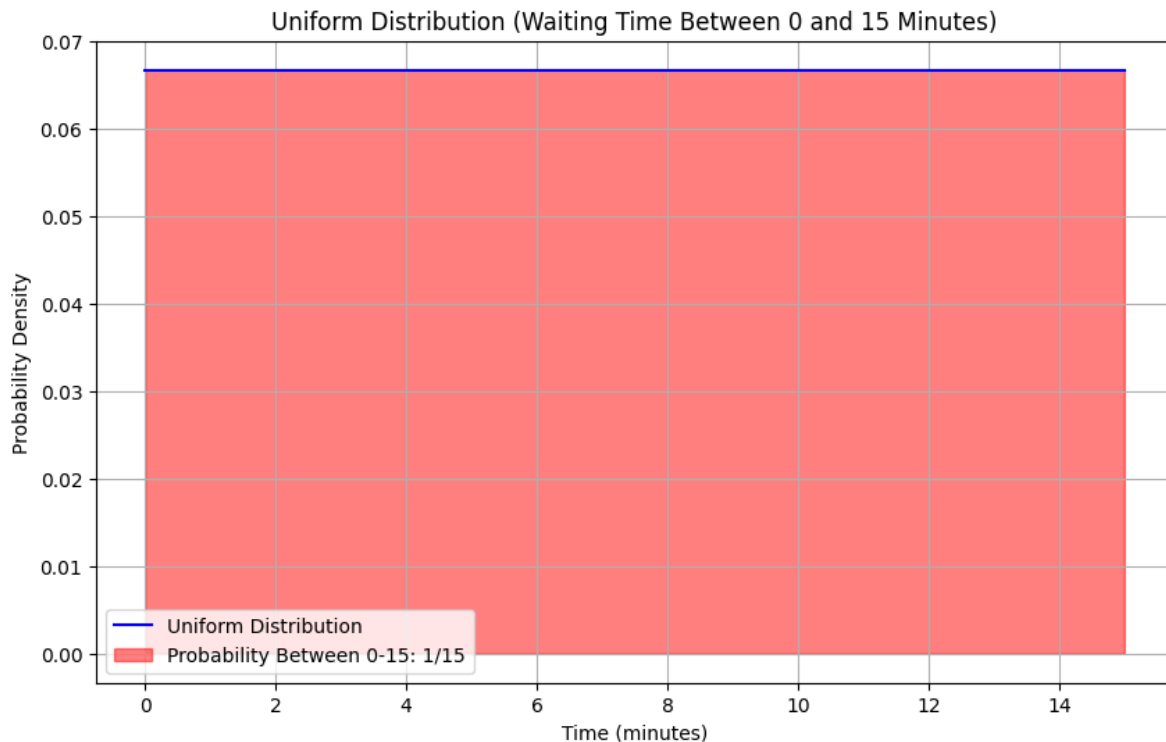
$$f(x) = \frac{1}{10-5} = \frac{1}{5} \quad \text{for } 5 \leq x \leq 10$$

$$f(x) = 0 \quad \text{otherwise}$$

In this case, the probability of the character's movement speed being any value between 5 m/s and 10 m/s is equal, with each value having a probability of 1/5

Graphical Representation

A uniform distribution graph shows a constant probability density over the interval [a,b]. The line will be horizontal and straight across the interval. Here is a simple example of a uniform distribution graph for bus waiting times between 0 and 15 minutes:



This graph shows that the probability of the bus arriving within 0 to 15 minutes is equal for each minute.

```
import numpy as np
import matplotlib.pyplot as plt
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```
# Uniform distribution parameters
a = 0
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Between 0-15: 1/15')

plt.title('Uniform Distribution (Waiting Time Between 0 and 15
Minutes)')

plt.xlabel('Time (minutes)')
plt.ylabel('Probability Density')
plt.legend()
plt.grid(True)
plt.show()
```

Conclusion

A uniform probability distribution is a type of distribution where all values within a specific range have an equal probability of occurring. This distribution is used to describe situations where any value within a range is equally likely to be chosen. Many real-world examples, such as drawing a card from a deck, tossing a coin, and waiting time for a bus, follow a uniform distribution. These concepts help STEM students understand the fundamental principles of probability and statistics.